

Race for Power in Public Good Games with Unequal, Unstable Punishment Power

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ABSTRACT

Decentralized punishment can maintain contributions to a public good at a high level. This positive effect holds even for situations involving inequality in punishment power, meaning that the relative costs for punishment differ between persons. We extend this line of research by examining the additional influence of instability of and thus potential competition for punishment power in public good experiments. As in previous research, inequality is operationalized by assigning the role of a 'strong player' to one group member, who can punish others at a reduced cost compared to other players. Instability is operationalized by the fact that the role assignment can change across rounds and is contingent on behavior in previous rounds. Three experiments show that under unequal, unstable (punishment-) power, individuals behave more competitively and engage in a *race for power* resulting in detrimental effects for the group. Specifically, we find that (i) unstable power goes along with a disproportionate use of punishment; (ii) not inequality by itself but rather the combination of instability and inequality causes this effect; and (iii) this excessive use of punishment is found not only when the role of the strong player is assigned to individuals who previously behaved rather selfishly but also when prosocial behavior leads to the top, speaking for a generalizability of our findings. Implications for organizational settings are discussed. Copyright © 2016 John Wiley & Sons, Ltd.

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The provision of public goods is essential for collectives, societies, and organizations. However, the non-excludability of individual group members from its consumption creates an incentive to free-ride on the contributions of others. It has been found that the opportunity to punish free-riders helps to increase individual contributions to the public good. Specifically, group members assign costly punishment to free-riders, which helps maintain contributions at a high level in repeated interactions. This positive effect also remains if inequality in punishment power exists, in that one specific group member can punish other team members at lower costs and therefore more effectively. What has not been investigated so far, however, is the role of instability in punishment power that often exists on top of inequality in real life situations. In such situations, the assignment of particularly high punishment power to one person in the group is not stable over repeated interactions but can change over time contingent on previous behavior. This makes punishment power obtainable for all group members. We argue that such situations bring about enhanced social comparison processes between group members resulting in a competition for power, which we refer to as the *race for power*. We investigate situations in which this *race for power* can have negative effects on group outcomes and discuss implications for organizations in which instable power hierarchies are relatively common.

Public goods and free-riding

Public goods are defined as goods that are shared among group members regardless of whether those members have contributed to the provision of the good or not (Olson, 1965). For example, a public good in the organizational context could be a group project whose outcome is rewarded with a bonus that is shared equally among all team members regardless of their individual contribution to the project's success. Situations like this give rise to a mixed-motive dilemma, wherein individuals must decide between maximizing their selfish interests, such as preferring to invest their time and resources elsewhere in the previous example, and maximizing collective interests, i.e., contributing as much as possible to the group project (see e.g., Komorita & Parks, 1995 and Van Lange et al., 2013, for reviews). Free-riding—i.e., benefiting from a public good without having contributed—is rational at the individual level (given that these actors aim to maximize their own outcome) but poses a serious problem for the group, because if everyone were to free-ride, there would be no public good to share among the group members, i.e., a group project would be unsuccessful in the above example. If everyone, however, were to contribute their full endowments, the collective outcome would be maximized.

Punishment and stable power

One way to handle the free-riding problem is to punish low-contributors (e.g., Fehr & Schmidt, 2000; Oliver, 1980; Ostrom et al., 1992; Yamagishi, 1986). It has been shown that people make use of the possibility of peer punishment even if it is costly and yields no material gain, which is why it is referred to as altruistic punishment (e.g., Fehr &

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Gächter, 2002). It has been shown that such punishment, directed to other group members for not having contributed to the provision of the public resource, has consistently produced higher levels of cooperation (e.g., Fehr & Gächter, 2002; Reuben & Riedl, 2013; see Balliet et al., 2011 and Chaudhuri, 2011, for reviews). Punishment can be implemented through various mechanisms that differ, for example, in terms of the individuals in the group who are allowed to punish. In the standard decentralized punishment design, punishments can be given by all group members (e.g., Fehr & Gächter, 2002); however, it is also possible to give only one member the power to punish (O'Gorman et al., 2009) resulting in a power inequality within the group. In the context of the group project, one could think of a situation where all group members have the opportunity to sanction people who did not contribute or alternatively one specific person in the group who has higher power and thus more punishment opportunities.

Previous research on public goods indicates that inequality in punishment power does not per se affect group outcomes (Nikiforakis et al., 2010) if the assignment of power is stable. Some research even indicates that in stable environments powerful people might use their power to the benefit of the group—i.e., to raise cooperation rates (O'Gorman et al., 2009).

Instability in power

In organizations and other real-life settings, however, assignments of power are often characterized by instability, in that a powerful person can lose his or her power, as in the case of short term contracts that might or might not be renewed, or performance-dependent project leadership. As a result, powerless individuals can possibly obtain and compete for power, whereas people in power might want to secure their position. We specifically address these kind of situations and investigate potential competition effects on group outcomes in situations characterized by unequal, unstable power.

From a theoretical and empirical perspective, a competition for power in unstable situations can be expected due to an automatic tendency rooted in social comparison tendencies. According to *Social Comparison Theory* (SCT, Festinger, 1954) people tend to compare themselves to others in social interactions in order to gain an understanding of their relative standing in the given domain. The aims of social comparison include ascertaining, improving, and/or enhancing one's own standing (Wood, 1996). When doing so, individuals primarily make upward comparisons (Wheeler, 1966); and followers engage in upward social comparisons with their leaders (Greenberg et al., 2007; Wood, 1989). Moreover, individuals also compare themselves with others who are in a worse position if they are threatened by a potential upward comparison (Garcia et al., 2006). Thus, competitiveness is one manifestation of the social comparison process (Garcia et al., 2013). Regardless of the direction of comparison, individuals are motivated to improve their relative position concerning a certain criterion only if a specific outcome (such as a powerful position) is obtainable (Lockwood & Kunda, 1997). These results suggest that in unstable as compared to

stable situations, individuals will compete with each other to obtain a better position within the group (in this case, to be the person in power).

As a result of competition for power, group members might be prepared to negatively affect other team members' chances of gaining or maintaining power instead of investing in group potentials (e.g., monetary resources or time). This in turn results in a waste of resources and negative outcomes for the group. Indeed, previous research has shown that instability of and potential competition for power can result in negative consequences for the group, in that individuals harm other group members instead of supporting group goals.

It has been shown that individuals whose power is being challenged behave more competitively to defend their position. Georgesen and Harris (2006), for example, found that individuals in such a situation award the least potential prize money to subordinates. Similarly, Maner and Mead (2010) show that powerful individuals who could potentially lose their power tend to defend their position by withholding information from other group members in a joint problem-solving task. On the other hand, in nonhuman primate groups, subordinates consistently challenge powerful group members when hierarchies are unstable, leading to an enhanced stress level on the part of the high-ranking individual (Sapolsky, 2005).

The effect of inequality and instability concerning punishment power in public goods has not yet been investigated. Consequently, even the most comprehensive models for behavior in public goods (e.g., Bogaert et al., 2008; Parks et al., 2013) do not cover such complex effects of competition. Based on the argumentation above, we suggest that, in the context of a public good, social comparisons could concern material outcomes but also non-pecuniary outcomes such as having more power to punish others. When levels of power differ among group members, but it is not possible for low-power individuals to obtain a high-power position, competition is unlikely to occur because it cannot be successful. However, if low-power individuals can obtain the high-power position, competition may occur, with low-power individuals investing resources to weaken the chances of other players to gain or maintain power, whereas the powerful person is expected to attempt to retain the powerful position by using every possible means. This competition for power is therefore expected to result in negative effects for the group. More specifically, we predict that instability concerning unequal punishment power leads to increased competition among group members—which we refer to as a *race for power*. For the introductory example, one could assume that team members attempt to obtain project leadership by overthrowing the current leader and negatively affecting other team members' chances to reach the top. These behaviors might come at the expense of the project outcome.

Unlike previous research on inequality in punishment power, we furthermore investigate effects of individual differences on contribution and punishment behavior, in general, and in unequal, unstable situations, in particular. One individual-difference construct that may be particularly relevant to one's engagement in competitive behaviors is

achievement motivation. Achievement motivation can be understood as the tendency to work towards a specific goal in a social environment (Cassidy & Lynn, 1989). More specifically, we included a measure of four achievement motives (*Achievement Motivation Scale*; Cassidy & Lynn, 1989) that can be assumed to be particularly relevant for situations involving social comparisons and competition for power. The four relevant subscales measure individuals' tendencies to strive for power and control over others (Dominance), social status (Status Aspiration), superior performance to others (Competitiveness), and monetary gain (Acquisitiveness for money).

Overview of the experiments

The current research investigates a potential competition for (punishment-) power in public good games with unstable power assignments. Experiment 1 thereby compares behavior in a public good game in which punishment power is unequal and unstable with the standard paradigm in which punishment power does not differ between group members (equal, stable). Experiment 2 serves as a broader replication of Experiment 1 and additionally includes a condition that is characterized by both inequality and stability in terms of (punishment-) power (unequal, stable). This additional condition was added to assess whether inequality per se or the combination of inequality and instability results in potential negative effects for the group. We investigate two different criteria for power assignment in unstable hierarchies to check for the generalizability of our results. Whereas in Experiments 1 and 2 the powerful position is assigned to the group member with the highest total income at the beginning of each round, the person with the highest social status (for details, see the description of Experiment 3) gains power in Experiment 3.

For all three experiments, we use pre-experimental questionnaires to measure the achievement motives mentioned above.

Experiment 1

Experiment 1 employs a strong manipulation comparing behavior in a public good game with equal (stable) punishment opportunities to a public good game with unequal, unstable punishment. As argued above, and based on previous findings indicating that the obtainability of a better standing within the group (in our case the high-power position) leads to processes of social comparison and thus higher competition within the group, we expect an enhanced competition in the latter. In this *race for power*, both low- and high-power individuals are expected to invest resources to weaken the chances of other players to gain or maintain power, which results in negative effects for the group.

Method

Participants and design. One hundred participants (mainly students from the University of Bonn, 46 female; 19–44 years of age) from the *MPI Decision Lab* subject pool

were recruited via the online recruitment tool Orsee (Greiner, 2004) and took part in the experiment. Participants played public good games in fixed groups of four. Sessions consisted of 12 and in some cases eight individuals due to occasionally high no-show rates. Assignment to groups was anonymous and random.

Participants were assigned to one of two conditions. They played either in an unequal, unstable condition (Table 1, Column 5)¹ or an equal, stable control condition (Table 1, Column 2). The condition was the same for all members of a group and all groups in a session. The experiment was computerized and run using Bonn Experimental System (*BoXS*, Seithe, 2012). The experiment consisted of an online pre-study to measure personality traits and a lab-study part conducted at least 12 hours later. Completing both parts of the experiment took participants overall approximately 90 to 120 minutes (pre-test approximately 30 minutes). Participants' payments depended on their decisions—earnings ranged from 17.10 to 29.70 Euros (approx. USD 22 to USD 39).²

Materials and procedures

Personality measures. The online pre-test measured four subscales of the Achievement Motivation Scale (AMS; Cassidy & Lynn, 1989).³ The four subscales were (i) Dominance, assessing a person's desire for power and authority; (ii) Status Aspiration, measuring the motivation to climb the social status hierarchy and desire to be a leader; (iii) Competitiveness, measuring the desire to compete with and outperform others in an activity; and (iv) Acquisitiveness (for money), measuring the motivation for material and monetary rewards. All items were answered on a 5-point scale (1 = strongly disagree, 5 = strongly agree). Scales were calculated so that high scores indicate high achievement motivations for the respective goal.

Public good game. Upon arrival participants read detailed instructions for the public good game and answered five to seven control questions (depending on the experimental condition) to test understanding. Experimenters checked answers and asked participants to reread the instruction and try again if they answered incorrectly. Participants were

¹To facilitate replication, all analysis and data files as well as all other materials (e.g., experimental instructions) related to the reported experiments will be made public under: osf.io/3qp7t.

²Additionally, participants played the opposite condition after a surprise restart. We observed strong carry-over and compensation effects from the first to the second part, which make the result difficult to interpret. We therefore report results from the first part only. Furthermore, in two groups of the control condition, the program had to be restarted at an early stage due to system break downs. We ran all analyses again without the critical groups; all results showed the same pattern as before (only the effect of condition on round win narrowly missed conventional significance levels—this might be due to lower statistical power when excluding these groups).

³In addition to these subscales, we used exploratory measures of risk- and loss aversion, testosterone (2D:4D), Social Value Orientation, Machiavellianism, and gender roles, not reported here.

Table 1. Overview experimental conditions in experiments 1 to 3

	(1) Equal punishment (no rank info)	(2) Equal punishment (rank info)	(3) Unequal, stable punishment	(4) Unequal, stable punishment (social)	(5) Unequal, unstable punishment	(6) Unequal, unstable punishment (social)
Punishment efficiency	$\bar{e} = 1.75$, $e_w = 1.75$	$\bar{e} = 1.75$, $e_w = 1.75$	$\bar{e} = 1.75$, $e_w = 1$ $e_s = 4$	$\bar{e} = 1.75$, $e_w = 1$ $e_s = 4$	$\bar{e} = 1.75$, $e_w = 1$ $e_s = 4$	$\bar{e} = 1.75$, $e_w = 1$ $e_s = 4$
Rank information	No	Income rank	Income rank	Social status	Income rank	Social status
Stable punishment	Yes	Yes	Yes	Yes	No	No
Unequal punishment	No	No	Yes	Yes	Yes	Yes
Determination of strong player	not applicable	not applicable	Randomly assigned (fixed)	Randomly assigned (fixed)	Highest total income	Highest social status
Experiment 1: 2; 5						
Experiment 2: 1; 2; 3; 5						
Experiment 3: 3; 4; 5; 6						

Notes: Overview of the experimental conditions. $\bar{e}$ indicates average punishment effectiveness; e_w and e_s indicate effectiveness for weak and strong players, respectively. In Experiment 1, the effectiveness of weak players and the average efficiency were both 1 in condition 2. The lower left-hand corner shows which conditions were implemented in the different experiments.

informed about the payment scheme and that communication among participants was forbidden throughout the experiment.

In the experiment, participants played 10 rounds of a four-person public good game (i.e., voluntary contribution mechanism) with costly punishment (e.g., Fehr & Gächter, 2002; Noussair & Tucker, 2005) in constant groups (i.e., fixed-partner design). Before the first round, participants received a lump sum of 250 Taler (experimental currency unit) to avoid absolute losses during the experiment. In each round, participants were given a round endowment of an additional 20 Taler (50 Taler = 1 Euro). In the first stage of each round, players decided simultaneously which amount between 0 and 20 of their round endowment to contribute to a group account. The remaining Taler were deposited to the respective player's private account. The amount of Taler contributed to the group account was multiplied by a factor of 1.6 (marginal per capita return $\alpha = 0.4$); the resulting amount was then equally divided between all four players of the group (and added to their private accounts). If all four participants of a group contributed their entire endowment of 20 Taler, each player would have earned an income of 32 Taler in this stage (i.e., 4 persons $\times$ 20 Taler $\times$ 1.6 / 4 persons = 32). Hence, there was a substantial potential collective gain of 60% compared to all individuals simply keeping their endowment and earning only 20 Taler.

In the second stage, participants could decide whether to apply costly punishment to other group members (in the instructions, we used the neutral term *reduction points*). Participants were informed about how much the other individuals in their group contributed in the current round. To ensure anonymity and avoid effects of gaining a reputation over rounds, players were presented by using numbers, which were randomly selected, ordered, and reshuffled in each round. As in previous research, inequality in punishment power was induced by manipulating the costs and thus the effectiveness of assigning reduction points between types of players (Nikiforakis et al., 2010). For the control condition with equal punishment, all players had a punishment effectiveness of 1. This means that they paid 1 Taler for reducing another player's income by 1 Taler (deducted Taler

were eliminated from the round). In the unequal, unstable condition, three of four players had the same punishment effectiveness of 1 as did all players in the control condition. One player, however, had a higher punishment effectiveness of 4 (i.e., each assigned reduction point deducted 4 Taler from the punishment target). Individuals with a punishment effectiveness of 1 were denoted weak players (in the instructions, we used the neutral term Type B players), whereas players with a punishment effectiveness of 4 were denoted strong players (called Type A players in the instructions). Participants were informed about power differences between group members in the unequal unstable condition. Instability was induced by giving each player the possibility to become the strong player. The maximum number of points a participant could distribute to each other player was 10 points for weak players and 5 points for strong players (resulting in $5 \times 4 = 20$ Taler damage). The latter limit for giving punishment was set to be lower than the former to avoid very high financial damages, especially for weak players, leading to absolute losses during the game. Such losses would make it impossible for weak players to reach the top position. Note that this limit works against our hypotheses, particularly in the analyses of the given punishment. Furthermore, this limit was reached in the minority of cases.⁴ Details regarding the experimental conditions in this and the following two experiments are provided in Table 1.

We chose prior performance in the public good as a plausible and meaningful basis for power assignment in the unequal, unstable condition. Thus, the strong punisher was the one with the highest total income at the beginning of each round (or by a random draw in case of ties between the persons with equal highest incomes).

Participants could gain an understanding of their relative standing within the group and furthermore estimate their chances to become the powerful person by the overall

⁴In Experiment 1, out of 420 individual punishment choices made by strong players, the 5-point limit was reached only 9 times. In Experiment 2, this limit was never reached. In Experiment 3, out of 482 individual punishment choices made by strong players, the 5-point limit was reached 34 times.

income ranking of all players. That is, in each round, they were informed which player earned most, second most, etc. in total over all previous rounds. Thus, participants were able to infer who was the strong and who were the weak players. To keep information constant across conditions, rank information regarding total income was also provided in the control condition.

At the end of each round, participants were informed about their total previous total income, their round income, their total income over rounds, the income reduction through punishment points they received, and the costs for assigned punishment points.

Furthermore, in the unequal, unstable condition, participants were informed about their player type (i.e., Type A [= strong] vs. Type B [= weak] player) for the next round but not about the player type of other players.

The round win for each player was their income after the first stage minus the costs for punishment points assigned to others and reduced by the amount deducted due to received punishments from all other players. The total income of a player is the sum of all of his or her round wins.

If the relative standing in a given domain, here punishment power, did not matter to individuals, they should (on average) invest in group resources equally in both conditions by contributing to the public good and engaging in altruistic punishment by punishing free-riders, as is the general finding in public good games. However, as argued above, we expect greater competition—a *race for power*—in the unequal, unstable condition as compared to the common public good game (equal, stable) leading to negative effects for the group. With the current mechanism by which power is assigned to individuals who compete for or attempt to maintain power, participants could potentially invest in their own interests instead of investing in group goals (i.e., free-ride, not invest in altruistic punishment) to have a high relative earning and thus a greater chance of being the powerful person. Furthermore, participants could attempt to lower the chances of other group members to become (continue to be) the powerful person by punishing other players—thus reducing their total income. Punishment that is not assigned in order to raise cooperation and stabilize high levels of cooperation rates (as is altruistic punishment) would be a waste of resources, because both group members suffer financial losses.

In summary, we expect an increased tendency to punish in the unequal, unstable condition as compared to the control condition when controlling for group contributions (H1a). Furthermore, we expect participants to make smaller contributions as compared to the control condition (H1b). More punishment and lower contributions should, in turn, result in lower group efficiency measured by the total amount of money that is earned by the individual participant in each round while taking into account both contributions as well as punishment (H1c).

Results. Overall, we collected data from 14 groups in the unequal, unstable condition and 11 groups in the equal, stable control condition. Due to repeated interactions within groups, observations within groups cannot be assumed to be independent. In the analyses, we corrected for resulting

dependencies in error terms by using regressions with cluster corrected standard errors over groups throughout the paper (Rogers, 1993).

To investigate our hypotheses, we compared average given punishment points, average contribution levels, and efficiency as indexed by average round win between the stable control condition and the unequal, unstable condition. Qualitatively, all effects are in the expected direction and appear to be stable over rounds. Punishment is consistently higher, whereas average contributions and round wins are lower in the unequal, unstable condition as compared to the control condition with equal, stable punishment (Figure 1).

To test hypotheses H1a to H1c statistically, we ran three OLS regressions predicting given punishment points, contributions, and round win by condition (dummy coded; stable (control)=1 vs. unequal, unstable punishment=0).⁵ For all regressions, we clustered at the level of groups and controlled for round effects and player type. In the regression analysis for investigating effects on punishment, we additionally controlled for the level of cooperation of the other players (i.e., sum of current round contribution of the other players in the group), because we were interested in the tendency to punish conditional on contributions.

Concerning punishment, there is a significant effect of our manipulation, thus supporting hypothesis H1a (Table 2, Column 1) that individuals assigned on average 0.89 fewer punishment points to other players in the equal, stable control condition as compared to the unstable situation (controlling for sum of other contributions).⁶

As can be gathered from Column 2 in Table 2, contributions do not differ significantly between the equal, stable punishment and the unequal, unstable punishment. Thus, the findings do not allow us to reject the null hypothesis in favor of H1b (i.e., that cooperation is reduced in unstable situations).

There is, however, a significant effect of condition on overall efficiency measured by individuals' average round win (Table 2, Column 3), which provides support for H1c. The average round win per individual was 5.12 Taler higher in the stable as compared to unstable condition (as indicated by the respective coefficient). According to the results reported above, this was driven mainly by differences in punishment.

We also investigated the behavior of strong players as compared to weak players. The regression results in Table 2 indicate that strong players contributed significantly less than weak players (i.e., 2.96 Taler). Players in the strong role did not assign more punishment points to other players than weak

⁵For two groups first round punishment was not recorded, thus punishments in these rounds were coded as missing values.

⁶For our analyses of contributions and round wins as well as for separate analyses of strong and weak players regarding punishment across conditions, we also conducted Tobit regressions taking into account the fact that ranges for these variables were artificially limited. These analyses led to the same results with respect to the conventional alpha error level of 5% (although there were some changes with respect to stricter alpha levels). Standard data truncation corrections (i.e., Tobit models) could, however, not be used to double check the robustness of the findings regarding overall punishment across conditions and punishment of strong as compared to weak players due to the different punishment limits of weak vs. strong players (5 vs. 10 points maximum). Thus, these results must be interpreted with caution.

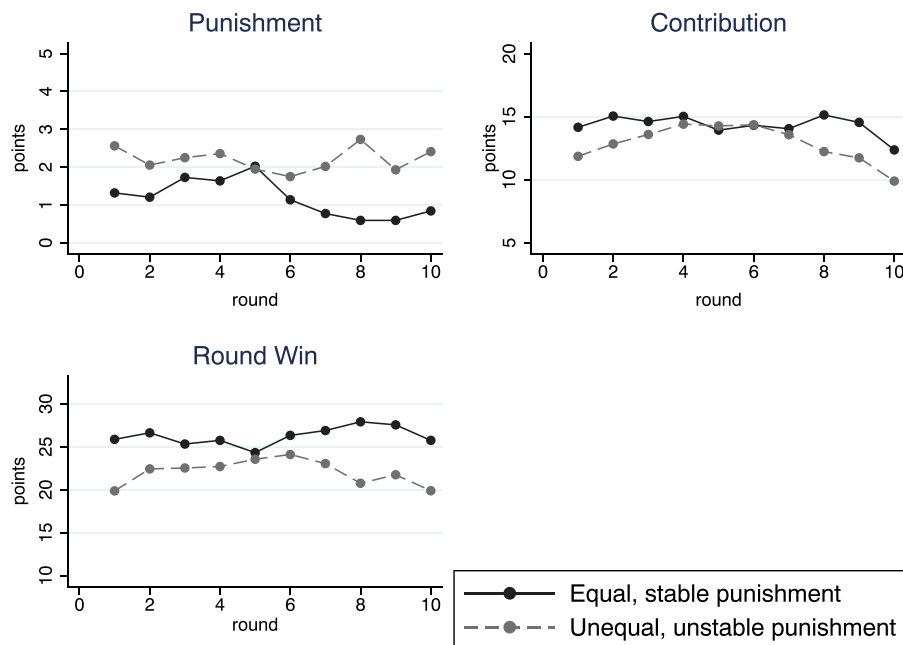


Figure 1. Punishments, contributions, round win by condition in Experiment 1. [Colour figure can be viewed at wileyonlinelibrary.com]

Table 2. Punishments, contributions, and round wins predicted by condition in experiment 1

	(1) Punishment	(2) Contribution	(3) Round win
Equal, stable (yes = 1, no = 0)	-0.893* (-2.13)	0.704 (0.39)	5.122* (2.51)
Strong player (yes = 1, no = 0)	-0.501 (-1.21)	-2.957** (-2.88)	3.942*** (4.87)
Sum of other contribution	-0.059*** (-4.29)		
Constant	5.057*** (6.42)	14.548*** (12.57)	21.027*** (12.10)
Observations	992	1000	992
Adjusted R^2	0.089	0.034	0.096

Note: *t*-statistics in parentheses, OLS regression analyses used, clustered at the level of groups, round dummies omitted. Sum of other contribution means the sum contributions of the three other players in a group in the current round. The unequal, unstable condition is the reference category.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

players but effectively caused more punishment damage ($b = 4.60$, $t(24) = 2.96$, $p = .007$), because every punishment point they assigned to other players was multiplied by four. Due to the lower costs of punishment and lower contribution rates, strong players earned significantly more points per round than weak players, $b = 3.94$, $t(24) = 4.87$, $p < .001$.

To investigate whether weak players behave more competitively in the unequal, unstable situation as compared to the stable situation, we compared the behavior of weak players between the two experimental conditions (due to the design, all players were weak players in the equal, stable condition). Our results reveal that, when controlling for the contribution of the other players, weak players assigned fewer punishment points to other players in the equal, stable condition as compared to the unequal, unstable condition,

$b = -0.88$, $t(24) = -2.07$, $p = .049$. Additional analyses including participant gender revealed no significant effects; and all reported results remain stable.

To shed more light on effects of individual differences on contribution and punishment behavior, we ran OLS regressions predicting given punishment points and contributions by the achievement motivations as well as condition and all two-way interactions between condition and personality factors. All four scales used to measure achievement motivations showed reasonable reliability and variability in our sample (desire for power and authority: Cronbachs $\alpha = .83$, $M = 3.42$, $SD = 0.67$); status aspiration: Cronbachs $\alpha = .76$, $M = 2.80$, $SD = 0.61$; competitiveness: Cronbachs $\alpha = .75$, $M = 2.16$, $SD = 0.65$; acquisitiveness for money: Cronbachs $\alpha = .77$, $M = 2.14$, $SD = 0.70$).⁷ When using a Bonferroni correction to account for multiple testing and the exploratory nature of the analyses (Bonferroni-corrected alpha level = $.05 / 4 = .0125$), none of results concerning personality factors reaches conventional significance.⁸

Discussion. Experiment 1 shows that individuals behave differently in an (equal) stable situation as compared to a situation characterized by inequality and instability in (punishment-) power. Even when controlling for the level of group cooperation, punishment was consistently higher in the unequal, unstable situation as compared to the control condition. This excessive use of punishment did not result in higher group cooperation but instead detrimental effects for the group in the form of lower group effectiveness as compared to the control condition. These results support the

⁷For all three studies, the theoretically expected 4 factor solution emerged when conducting a factor analysis using the *Kaiser Criterion*. Hence, the measured subscales appear to be largely independent. However, we observe some small deviations at the item level.

⁸The corresponding tables can be also found here: osf.io/3qp7t.

assumption of greater competition—a *race for power*—in the unequal, unstable condition, resulting in a waste of resources. This detrimental use of punishment complements previous research on antisocial and perverse punishment (Cinyabuguma et al., 2006; Herrmann et al., 2008), both of which describe comparatively high punishment towards high contributors.

Furthermore, our results reveal that especially weak players behave competitively in unequal, unstable situations. In sum, our results support the assumption of higher competition in unstable situations, which results in detrimental effects for the group. One could argue for other potential motivations behind the observed behavior such as rational money-maximizing concerns. However, as the powerful position was not rewarded with an additional monetary benefit in our experiment, the alternative explanation of rational agents is less likely.⁹ Furthermore, in order to maximize outcomes, one should refrain from wasting resources by punishing others heavily.

There are some limitations to the current study. First, although the comparison with the study by Nikiforakis et al. (2010) as well as the theoretical and empirical basis outlined in the introduction would suggest that instability leads to competition and, thus, negative group outcomes, our results cannot directly address the question whether the observed effects are driven by inequality, instability, or a combination of both factors. Second, average punishment effectiveness, which has been shown to increase contributions (Egas & Riedl, 2008; Nikiforakis et al., 2010), is higher in the unequal, unstable ($\bar{e}=1.75$) as compared to the equal condition ($\bar{e}=1$), potentially leading to lower levels of contributions in the control condition. Third, although we believe it was important to hold rank information constant across conditions in order to avoid a confound regarding competition for rank and power, it might be criticized that our control condition deviates from standard controls and might have influenced our findings in an unpredicted manner.

Experiment 2

Experiment 2 was designed to address the above mentioned methodological issues by including a between-subjects comparison with an unequal, stable punishment condition. Furthermore, we used a broader replication approach including an additional control condition without rank information, using the same average punishment effectiveness ($\bar{e}=1.75$) for all conditions. We expected effects found in Experiment

1 to be caused by instability in combination with inequality and not by inequality alone. Thus, we expected an increased tendency to punish in the unequal, unstable condition as compared to the unequal, stable condition when controlling for group contributions (H2a). Furthermore, we expected participants to make smaller contributions as compared to the control condition (H2b). More punishment and lower contributions should, in turn, result in lower group efficiency measured by the total amount of money that is earned by the individual participant in each round while taking into account both contributions and punishment (H2c).

Methods

Participants and design. One-hundred-twenty-eight participants (81 female, 45 male, 2 not reported; 18–35 years of age, 6 not reported) participated in the experiment and were recruited and assigned to groups according to the same protocol used in the first study. Participants' payments again depended on their decisions and earnings, ranging from 16.00 to 21.60 Euros (approx. USD 20 to USD 29).¹⁰ Participants were assigned to one of four conditions, which are summarized in Table 1 (columns 1, 2, 3, and 5). The previously used unequal, unstable condition and the equal, stable control condition with rank information (equal, rank info) were supplemented with an additional unequal, stable condition and an additional control condition without information concerning current ranking (equal, no rank info).

Personality measures. In an online pre-test, we again measured the four subscales of the Achievement Motivation Scale, namely Dominance (Cronbachs $\alpha=.78$, $M=3.46$, $SD=0.59$), Status Aspiration (Cronbachs $\alpha=.70$, $M=2.81$, $SD=0.54$), Competitiveness (Cronbachs $\alpha=.82$, $M=2.39$, $SD=0.73$), and Acquisitiveness (Cronbachs $\alpha=.77$, $M=2.19$, $SD=0.69$).

Public good game. The unequal, stable condition was implemented in that the role of the strong player was assigned randomly at the beginning of the first round and could not change from one player to another during the game. The new equal, no rank information control condition involved a standard public good game with punishment, in which the rank information from the previous control condition was not provided. In contrast to the first experiment, the mean punishment effectiveness was held constant across all conditions and set to $\bar{e}=1.75$. Note that it was not possible to apply a full factorial design here, because instability presupposes inequality in punishment power in the first place. For pragmatic reasons, we did not fully cross the rank-info manipulation with the other factors.

Results. In line with our second set of hypotheses, punishment was consistently higher, whereas contributions and round wins were lower in the unequal, unstable condition as compared to the unequal, stable condition (Figure 2).

⁹From the classic economic perspective, a money-maximizing agent would never make use of costly punishment in the last of several interactions, regardless of whether he or she is a powerful or powerless player. Other rational players would anticipate this behavior and, thus, refrain from contributing to the second-order (and the first-order) public good in the penultimate round and so forth. Instability in punishment power would not change this game-theoretical solution provided that achieving power is not rewarded with an additional monetary benefit. We focus on the standard solution concepts for dynamic games with observable actions under the standard textbook assumptions of self-interest and common knowledge of rationality. The unique subgame-perfect equilibrium can be derived through reversed induction.

¹⁰We lost data from four groups due to system failures.

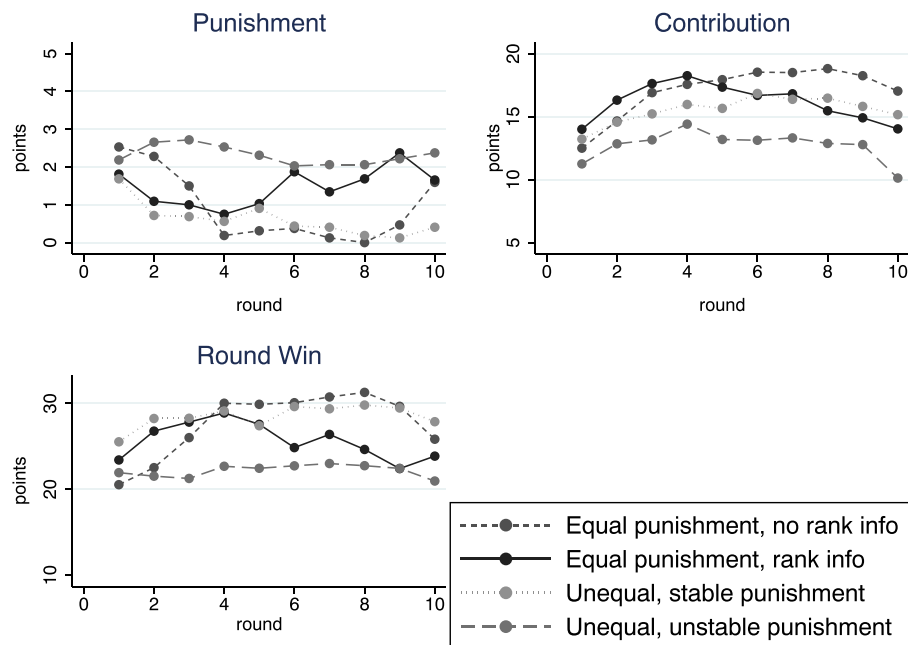


Figure 2. Punishments, contributions, round win by condition in Experiment 2. [Colour figure can be viewed at wileyonlinelibrary.com]

Hence, there is descriptive support for the notion that the combination of inequality and instability instead of inequality alone led to the effects observed in Experiment 1.

To test hypotheses H2a to H2c statistically, we ran three separate OLS regressions comparing given punishment points, contributions, and round wins in the unequal, unstable condition compared to the unequal, stable condition and the two control conditions. All conditions were included as dummy codes comparing against the unequal, unstable condition as the reference group and controlling for the sum of other contribution and player type (Table 3).

Concerning punishments, there is a significant effect of instability such that when holding the level of group cooperation constant, fewer punishment points were assigned to other players in the unequal, stable condition than in the unstable counterpart, thus supporting hypothesis H2a (Table 3, column 1). Although there is a tendency in the expected direction, contributions in the unequal, stable condition were not significantly larger than in the unstable condition (Table 3, column 2). Hence, the findings do not allow us to reject the null hypothesis in favor of H2b. Round wins were higher in the unequal, stable condition than in the unequal, unstable condition (Table 3, column 3), which supports H2c. The general hypothesis that the combination of instability and inequality is crucial for the detrimental effects found in Experiment 1 is again supported in Experiment 2.

Overall, players in the strong player role assigned significantly fewer punishment points to other players, contributed less than individuals in the weak player role, and earned significantly more per round (Table 3).

To investigate whether strong players' behavior varied with the stability of the punishment assignment, we compared the behavior of strong players between the two unequal (stable vs. unstable) conditions. Interestingly, punishment by strong players does not differ significantly when comparing

the stable and unstable conditions, $b = -0.49$, $t(15) = -1.18$, $p = .26$. The same is true for contribution behavior, $b = 3.21$, $t(15) = 1.07$, $p = .30$. Concerning round win, strong players suffered from instability, in that their average round win (in the rounds in which they were the strong players) was significantly higher in the stable as compared to the unequal, unstable condition, $b = 5.44$, $t(15) = 2.30$, $p = .04$.

Holding group cooperation constant, weak players punished significantly less in the stable condition as compared to the unequal, unstable condition, $b = -1.45$, $t(15) = -2.38$, $p = .03$. Their contributions did not differ

Table 3. Punishments, contributions, and round wins predicted by condition in experiment 2

	(1) Punishment	(2) Contribution	(3) Round win
Equal, no info (yes = 1, no = 0)	-0.781 (-1.34)	4.009 ⁺ (1.93)	5.924* (2.05)
Equal, rank info (yes = 1, no = 0)	-0.432 (-0.72)	3.084 (1.64)	3.925 (1.35)
Unequal, stable (yes = 1, no = 0)	-1.083* (-2.19)	2.822 (1.19)	6.300* (2.30)
Strong player (yes = 1, no = 0)	-0.871** (-2.94)	-1.415* (-2.16)	1.743* (2.44)
Sum of other contribution	-0.063*** (-3.90)		
Constant	5.200*** (5.48)	12.451*** (7.82)	20.737*** (7.90)
Observations	1,280	1,280	1,280
Adjusted R^2	0.158	0.072	0.114

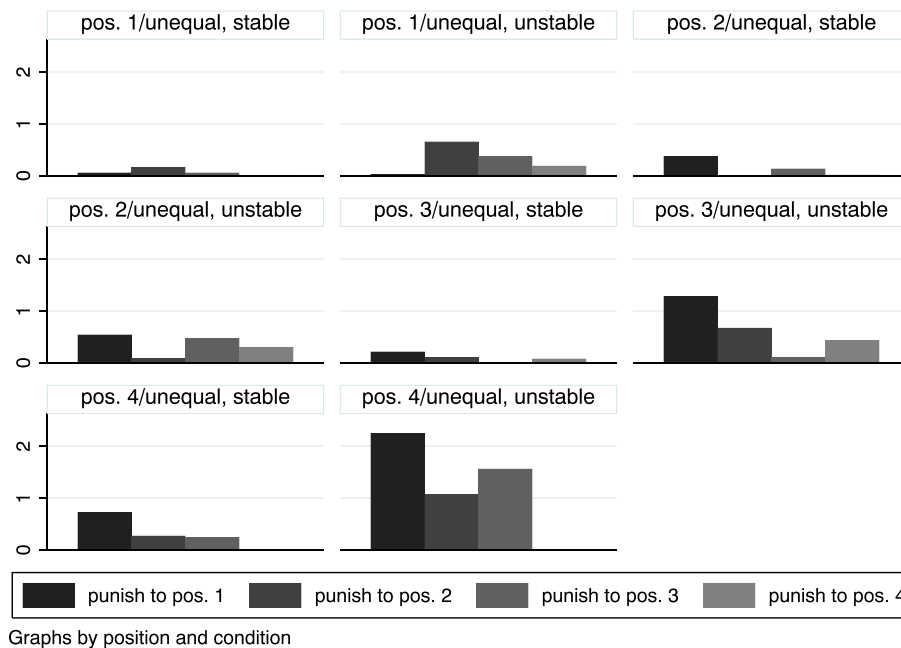
Note: t -statistics in parentheses, OLS regression analyses used, clustered at the level of groups, round dummies omitted. Sum of other contribution means the sum contributions of the three other players in the current round. The unequal, unstable condition is the reference category.

⁺ $p < 0.1$.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.



Graphs by position and condition

Figure 3. Punishments to and by position for the unequal, stable and the unequal, unstable condition in Experiment 2 (without first round in which all participants are in position 1). Because ranks are shared in case of ties in total income, punishing the own rank is possible. [Colour figure can be viewed at wileyonlinelibrary.com]

significantly between conditions, $b = 2.69$, $t(15) = 1.18$, $p = .26$. In addition, they earned significantly more in the stable condition, $b = 6.59$, $t(15) = 2.20$, $p = .04$. This indicates that weak players reacted quite differently to their roles in stable versus unstable environments in that they behaved competitively in the latter by assigning more punishment points to other players than in the stable condition, thus replicating the results of Experiment 1.

Figure 3 further illustrates that not only the strong players but even more so the weak players assigned more punishment points in the unequal, unstable condition as compared to the unequal, stable condition.

Regarding personality measures, the only main effect that reaches significance in the analysis over all conditions is the effect of Acquisitiveness (for money) on contributions, $b = -2.19$, $t(31) = -5.17$, $p < .001$ (Bonferroni-corrected alpha level = $.05 / 4 = .0125$). That is, participants contributed less with increasing motivation for earning money.

DISCUSSION

Experiment 2 provides important additional support for the hypothesis that inequality and instability lead to enhanced competition due to the obtainability of a better standing in the group. Subordinates, in particular, engaged in greater competition in such situations, resulting in lower group outcomes in the unstable as compared to the stable power hierarchy. Beyond a replication of the detrimental effects of instability on group outcomes, this study additionally shows that the detrimental effects are driven by a combination of inequality and instability in punishment and not by inequality in punishment per se as indicated by the comparison to the unequal, stable condition, which is consistent with Nikiforakis et al.'s (2010) findings.

In Experiment 2, the mechanism through which the strong player was selected favored free-riders, because individuals who contributed little would have been more likely to have higher total incomes than high-contributors and, as a result, be more likely to become strong players. Free-riders, by definition, contribute less than the group average and may be more likely to punish others. Furthermore, one could argue that our results are highly context-specific and that the problem might be solved when prosocial rather than free-riding behavior leads to the top. Thus, we cannot rule out the possibility that the selection mechanism favoring free-riders, rather than instability per se, led to the observed results. Moreover, in real-life group settings, individual performance is certainly not the only criteria for promotion to high power positions—social aspects are also taken into account.

To address this issue, we changed the selection mechanism in Experiment 3, so that players who behaved in favor of the group (i.e., individuals who behaved prosocially and contributed a relatively high share to the public account) were more likely to be rewarded with the powerful position. If the observed effects are a result of selection effects, the adverse effects of unstable power asymmetries should disappear. However, we argue that the negative effects of social comparison persist, because subordinates can still compare themselves to others in terms of power and, as a result, compete for the powerful position.

Experiment 3

Experiment 3 was designed to explore whether stronger competition in unequal, unstable situations due to processes of social comparison—a *race for power*—can also be observed when the selection mechanism does not favor selfish (anti-social) behavior. As the strongest test, we implemented a selection mechanism that promoted prosocial behavior - the

strong player role was assigned to the person with the highest social status (i.e., the sum of contribution minus received punishments). We expected the enhanced competition found in Experiment 1 and 2 to persist in this environment, even though becoming the powerful person was compatible with group goals. Individuals could enhance their chances to become the strong player by contributing their resources. At the same time, punishment can still be misused to negatively affect the potential of other group members. In detail, we again expected higher punishment in the unstable situation as compared to the stable situation (H3a). Regarding contributions, however, we expected larger contributions in the latter (H3b), because making large contributions is the other behavioral option to succeed in the competition (i.e., gain punishment power). These two effects work in opposite directions and can be expected to cancel each other out and result in no differences concerning group efficiency (H3c) in the unequal, unstable punishment relative to the unequal, stable condition.

Methods

Participants and design. One-hundred-ninety-two participants (115 female, 75 male, 2 not reported; 18–72 years of age) took part in the experiment and were recruited and assigned to groups according to the same protocol used in the other two experiments. Participants' payments again depended on their decisions; and earnings ranged from 11.60 to 17.50 Euros (approx. USD 15.90 to USD 21.60). Participants were assigned to one of four conditions. The previously used unequal, unstable condition and the stable counterpart were again used to implement the manipulations with anti-social favoring selection mechanisms. Additionally, we added two new conditions with a new prosocial favoring selection mechanism, which is explained below (Table 1, columns 3–6).

Personality measures. In an online pre-test, we again measured the four subscales of the Achievement Motivation Scale, namely Dominance ($\alpha = .85$, $M = 3.31$, $SD = 0.72$), Status Aspiration ($\alpha = .77$, $M = 2.67$, $SD = 0.66$), Competitiveness ($\alpha = .77$, $M = 2.13$, $SD = 0.68$), and Acquisitiveness ($\alpha = .81$, $M = 1.96$, $SD = 0.75$).

Public good game. In the lab, we implemented the new selection mechanism, which was based on accumulated contributions minus accumulated received punishments instead of total incomes (Table 1, column 6). In doing so, we changed the selection mechanism such that prosocial rather than selfish (anti-social) behavior was rewarded with power (unequal, unstable (social)), subsequently ruling out the possibility of selection effects. To be more precise, in each round the role of the strong player was assigned to the player with the highest total social status (total contributions minus total punishments). Consequently, the provided information on current ranks was based on the total social status.

In order to again compare the stable and unstable situations, we added the corresponding stable counterpart (unequal, stable (social)), in which the strong player role was randomly assigned prior to the first round and held constant (Table 1, column 4). This condition differed from the previous unequal, stable condition used in Experiment 2, in that the information on current ranks was based on the total social status rather than total incomes.

Results. In line with our third set of hypotheses, when considering the two conditions with the prosocial favoring selection mechanism only, contributions and given punishments were consistently higher in the unstable, social condition, whereas there was no difference in round wins (Figure 4). Hence, there is descriptive support for the notion

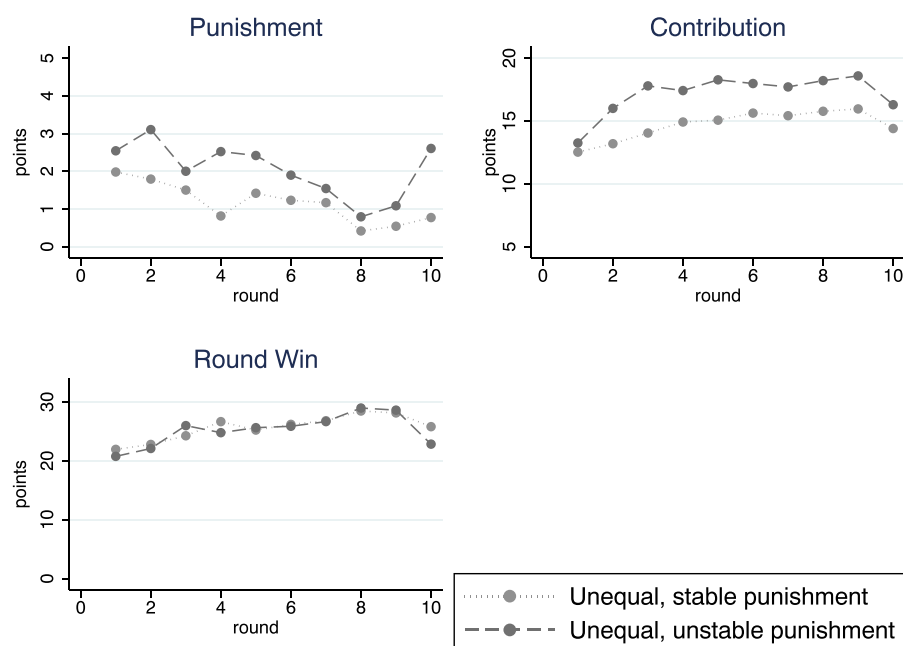


Figure 4. Punishments, contributions, round win by (social) condition in Experiment 3. [Colour figure can be viewed at wileyonlinelibrary.com]

Table 4. Punishments, contributions, and round wins predicted by condition in experiment 3 (social conditions)

	(1) Punishment	(2) Contribution	(3) Round Win
Unequal, stable (yes = 1, no = 0)	−1.296* (−2.62)	−2.460 (−1.67)	0.410 (0.23)
Strong player (yes = 1, no = 0)	−0.341 (−1.13)	0.318 (0.59)	1.460** (3.34)
Sum of other contribution	−0.055* (−3.21)		
Constant	5.502*** (6.24)	15.484*** (17.26)	21.834*** (13.35)
Observations	960	960	960
Adjusted R^2	0.088	0.057	0.053

Note: *t*-statistics in parentheses, OLS regression analyses used, clustered at the level of groups, round dummies omitted. Sum of other contribution means the sum contributions of the three other players in the current round. The unequal, unstable condition is the reference category.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

that competition is also enhanced in an environment that fosters prosocial selection.

To test hypotheses H3a, H3b, and H3c statistically, we ran three separate OLS regressions comparing given punishment (controlled for sum of other contribution), contributions, and round wins in the social, unequal, unstable condition compared to the social, unequal, stable condition (Table 4).¹¹ A lower level of punishment could be observed in the stable as compared to the unstable condition, thus supporting H3a (Table 4, column 1). Furthermore, contributions in the stable condition were lower than in the unstable counterpart (Table 4, column 2). However, this result does not reach conventional levels of significance.

Round wins were almost equal in both conditions (Table 4, column 3), providing some support for the null hypothesis H3c. Power estimations for cluster regressions are difficult to calculate and the results depend strongly on assumptions concerning dependencies of observations—our estimates of power range from .13 to .75.¹² In both cases, the result would only provide weak to moderate support for the null hypothesis H3c.

For the two conditions involving the anti-social favoring selection mechanism (unequal, unstable vs. unequal, stable with the previously used selection mechanism), the effects are in the same direction as in Experiment 2 but do not reach significance (Table 5).

Further analyses of the conditions involving a social selection mechanism (social conditions) indicate that persons

Table 5. Punishments, contributions, and round wins predicted by condition in experiment 3 (antisocial conditions)

	(1) Punishment	(2) Contribution	(3) Round win
Unequal, stable (yes = 1, no = 0)	−0.208 (−0.39)	0.811 (0.40)	0.279 (0.13)
Strong player (yes = 1, no = 0)	0.585 (0.90)	−0.798 (−1.67)	1.905*** (3.94)
Sum of other contribution	−0.006 (−0.36)		
Constant	1.958** (2.88)	14.177*** (10.83)	23.218*** (14.57)
Observations	960	960	960
Adjusted R^2	0.003	0.003	0.011

Note: *t*-statistics in parentheses, OLS regression analyses used, clustered at the level of groups, round dummies omitted. Sum of other contribution means the sum contributions of the three other players in the current round. The unequal, unstable condition is the reference category.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

in the strong player role did not differ significantly in terms of punishments and contributions but did earn significantly more per round than weak players (Table 4).

Strong players contributed significantly less in the stable situation as compared to the unstable situation, $b = -4.34$, $t(23) = -2.28$, $p = .03$. Hence, there is a positive effect of instability, at least for strong players. This finding is in line with results by Sligte et al. (2011), who found that subordinates engage in more creative effort when the powerful position is obtainable, especially when being creative is relevant to power. As in Experiment 2, weak players punished significantly less in the stable condition as compared to the unequal, unstable condition, $b = -1.48$, $t(23) = -2.71$, $p = .01$. Figure 5 shows assigned punishment points broken down by position and experimental condition. Players in low positions assigned more punishment points in the unstable as compared to the stable environment. Thus, punishment was directed at both players with a high and low social status.

This indicates that weak players again behave competitively when the situation is characterized by a combination of inequality and instability and were mainly responsible for the negative outcomes for the group. As in Experiment 1, none of the results concerning personality factors reach conventional significance. Also, no gender effects were observed.

Discussion. Experiment 3 provides additional support for the hypothesis that the combination of inequality and instability leads to a *race for power*. We show that detrimental effects can also be observed when selection procedures favor prosocial rather than selfish behavior. In Experiment 3, the detrimental effects on efficiency are not that obvious at first due to the existence of counteracting effects that cancel out each other. Group efficiency (average round win) is highly similar in both situations. However, with regard to the cooperation rate, it becomes obvious that the groups could have performed better if participants did not waste resources to gain a better relative standing in the group—i.e., to achieve the powerful position.

¹¹Due to a programming error, in a few trials (1.3%) two persons with the same criterion value were assigned the role of a strong player at the same time. Excluding these trials or the whole groups from the analyses, however, did not change the results.

¹²Assuming a small to medium effect size $d = .35$ based on previous work (e.g., Maner & Mead, 2010, Table 1) and independence of repeated measures for groups ($N = 23$ groups $\times$ 10 rounds = 230) results in sufficient power = .75 for the two-sided test (Faul et al., 2007). Note, however, that the power of the test drops to .13 when assuming complete dependence over time and only $N = 23$ groups.

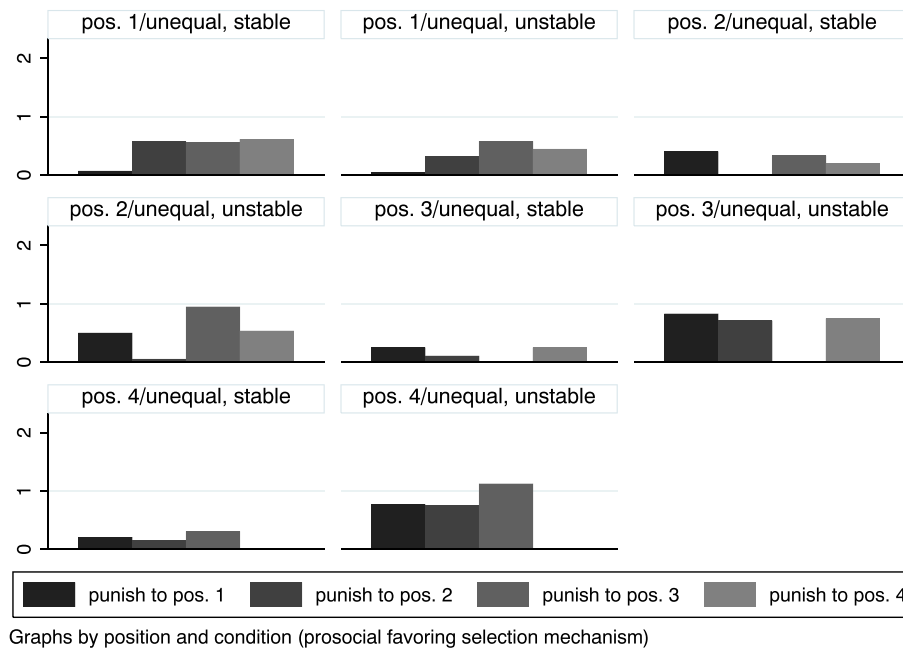


Figure 5. Punishments to and by position for the social unequal, stable and the social unequal, unstable condition in Experiment 3 (without first round in which all participants are in position 1). Because ranks are shared in case of ties in total income, punishing the own rank is possible. [Colour figure can be viewed at wileyonlinelibrary.com]

General discussion

Previous research shows that decentralized costly punishment consistently improves group outcomes by stabilizing cooperation rates in public good games at a high level (see Balliet et al., 2011, for a meta-analysis). This positive effect of punishment remains stable when punishment power is distributed asymmetrically among group members (Nikiforakis et al., 2010). In the current research, we show a negative side-effect of punishment opportunities in situations in which inequality in punishment power is complemented with instability, in that strong players can lose their position and weak players can gain punishment power based on their decisions.

In three experiments, we demonstrate that the combination of inequality and instability leads to a *race for power*—i.e., stronger competition among group members. This competition becomes evident in individuals not only investing in group potentials but also negatively affecting the potentials of other group members, i.e., their chances of becoming the strong player. They use punishment opportunities not only to raise or stabilize contribution levels but also to keep other players down, resulting in a waste of resources, which in turn reduces group efficiency. This finding is in line with our predictions derived from *Social Comparison Theory* (Festinger, 1954) and previous empirical results indicating that the powerful position itself can become the object of competition if instability prevails. In other words, when weak players have the opportunity to obtain the powerful position, as is the case in an unstable hierarchy, they may begin to compete with each other as well as with the strong player in order to obtain the powerful position. In contrast, when the powerful position is stable, there is nothing to compete for; thus, losses due to competition can be avoided.

In line with this theoretical account, instability of a powerful position leads to detrimental effects in social

dilemmas due to individuals striving to achieve the powerful position. They appear to do so by using whatever behavioral option is available. In situations in which the selection mechanism favors selfish behavior, two cumulating sources cause detrimental effects: individuals punish others to sweep them out of the way and also refrain from contributing their resources to the public account. However, even in more social environments in which selection mechanisms favor prosocial behavior, individuals misuse the opportunity to punish in order to get to the top. More detailed analyses of the behavior of the strong and weak players revealed that weak players, in particular, behave more competitively in the unstable environments than in their stable counterparts.

Limitations and future directions

One limitation is that, although Experiment 3 showed the predicted results for the new conditions with the prosocial selection mechanism, the results concerning the anti-social selection mechanism, although descriptively as expected, did not reach conventional levels of significance. As has been repeatedly highlighted in recent methodological debates, one may not expect all effects to replicate all the time, as this is statistically very unlikely (e.g., Francis, 2012). Given that the expected effect was consistently observed in the first two experiments and was consistently in the expected direction as well as the fact that we also observed the effect for the prosocial conditions in the same experiment, we would argue that the lack of a significant finding is most likely due to random fluctuation. Closer inspection of the data from Experiment 3 (Table 5) supports this explanation by revealing that behavior in the anti-social selection mechanisms was generally unusual in that, in contrast to all other studies, punishment did not increase with

lower contributions of others in the current round, which is counterintuitive. Still, future studies might be conducted to provide additional replications to rule out any remaining concerns.

As a second limitation, it might be noted that the investigation of personality factors reported in this manuscript was largely exploratory and only one result reached significance when applying a Bonferroni correction to account for multiple testing. Furthermore, this effect (Acquisitiveness for money on contribution) was found only in one of the studies. Thus, further studies are necessary to provide more clear-cut results concerning the effect of personality on the *race for power*.

Third, for pragmatic reasons, we had to limit our investigation to several specific selection mechanisms. Future investigations might extend this approach and could also test whether the effects hold when additional properties of the selection mechanism are varied. For example, it may be useful to select the strong player based on some desirable characteristics such as intelligence, ability, or reputation.

Despite some limitations, our research provides interesting routes for future research. We argue that processes of social comparison and the desire for a better standing within the group are responsible for enhanced competition and the negative effects we observed in our experiments. Two other observations would support this assumption: When examining Figures 3 and 5 more closely, it becomes evident that punishment was not only directed at the person in power but also substantially to adjacent upper and lower positions, which is in line with Wheeler (1966) and Garcia et al. (2013). Second, Experiment 2 demonstrated that a (minor) *race for power* can also be induced by an even more subtle manipulation such as whether or not individuals are provided with internal rank order information. When rank-information is provided, *Social Comparison Theory* (Festinger, 1954) would also predict that individuals will begin to compete for a better position relative to others. It appears promising to investigate the effects of enhanced social comparison when there are several domains determining the relative standing within a group. Second, although we consistently used the term “*race for power*”, another term might be equally applicable: A “*race away from a lack of power*”. The fact that we also observed enhanced punishment in the equal, stable condition with rank information as compared to the standard paradigm (equal, stable, no rank information) would suggest that individuals care about their relative standing in public good games also apart from a potential fear of powerlessness. However, our data does not allow us to sufficiently distinguish between these two motivations. Future research could, for example, investigate approach and avoidance tendencies (e.g., Elliot & Covington, 2001) to learn more about the underlying motivations.

Practical implications

Our findings allow us to cautiously derive recommendations for organizational structures. They suggest that environments that combine inequality and instability but also involve interdependence among group members could result in a waste of

valuable resources. Instead of pursuing group goals, individuals might invest time and/or financial resources to keep potential rivals down. The consequences of such behavior might mean that organizations cannot exploit their maximum potential, suffer from financial losses, or might not be able to withstand competition with other organizations. One obvious solution to this problem is to not bring powerful people into unstable situations by, for instance, providing them with short term contracts and thereby enabling their staff members to take their position. Instead, one might instead ensure that the group leader has sufficient security to keep his or her position. However, with regard to recent developments, it appears unlikely that short-term positions can or will be avoided within the near future. Thus, one should instead concentrate on compensatory incentives that encourage cooperative motivations under such circumstances. Our experimental results show that when assignment of power is based on behavior that is beneficial to the group, positive effects of competitions (i.e., higher cooperation) could also be observed. This would suggest that selection criteria for promotion should also include social skills. However, in such situations we also found that individuals are still willing to harm their group members, paradoxically resulting in pro- and antisocial behaviors at the same time. One way to overcome such effects is to also take into account the degree to which potential candidates for leadership positions supported their teammates in the past. In addition, the establishment of sufficient monitoring institutions to detect harmful action against colleagues might be a reasonable strategy.

APPENDIX:

A. Sample instructions for equal punishment, no rank info

You are about to take part in an economic experiment. In this experiment you can earn a significant amount of money depending on the choices you make. For this reason, it is essential to read the instructions carefully.

The instructions you obtained are for your private use only. Please note that you are not allowed to communicate with other participants during this experiment. Should you have any questions, please ask us for assistance. If you do not comply with the rules, we will have to exclude you from the experiment and from all payments.

During the experiment, we do not speak of euros but of Talers. Your income will be calculated first in Talers. The total number of Talers collected during the experiment will be converted to euros and rounded to one decimal place according to the following rule:

$$50 \text{ Talers} = 1 \text{ Euro}$$

At the beginning, every participant receives a one-time lump sum payment of **250 Talers**. With these 250 Talers you can make up for possible losses. The total profit made

during the experiment will be paid out in cash at the end of the experiment.

Today's experiment consists of several parts. This part of the experiment consists of **10 rounds** in total. Participants are randomly assigned to groups of 4. Thus, apart from you, your group has **3 more players**. For these 10 rounds the composition of your group remains unchanged. Therefore you are in a group with the same persons for 10 rounds. Please note that the identification numbers assigned to you and the other players of your group change randomly. In this way, the individual players cannot be identified over the 10 rounds.

Every round consists of **2 stages**. In the first stage you have to decide how many Talers you want to contribute to a project. In the second stage you are informed about the contributions of all other players and by assigning reduction points, you can decide whether and how much the income of the other players is reduced.

On the following pages we describe the exact procedure of the experiment.

Information on the exact procedure of the experiment

The first stage

At the beginning of each round every participant receives **20 Talers**, which we refer to as the player's **endowment** in the following. Your task is to make a decision regarding the use of your endowment. You have to decide how many of the 20 Talers you contribute to a group project and how many of them you keep for yourself. The consequences of this decision will be explained below in more detail.

In this round you receive an endowment of **20 Talers**.
Please decide how much of that endowment you want to contribute.
Your contribution to the group project is:

In each round your **endowment is 20 Talers**. You make a decision about your contribution by entering a number between 0 and 20 in the input field on the screen. By deciding on your contribution, you automatically decide how many Talers you keep for yourself, namely **20 minus your contribution**.

After all players of your group have made their decision, you are informed about the sum of all contributions (including yours) which your group made to the project. Additionally, you are informed about your own contribution, the number of Talers you kept for yourself, as well as your total income from the first stage.

Your income from the first stage consists of two parts:

- (1) The Talers, you kept for yourself ("income from Talers kept")
- (2) and the "income from the project". Your income from the project is 0.4 times the sum of all contributions to the project.

Thus your income from the first stage is calculated as follows:

Income at the end of the first stage :

= Income from Talers kept + Income from the project where

Income from the project = $0.4 \times \text{sum of all contributions to the project}$

The income at the end of the first stage is calculated for all players of your group according to the same formula.

If, for example, the sum of all contributions of the group members is 60 Talers, you and the other players from your group receive the income from the project, equal to $0.4 \times 60 = 24$ Talers. If the players contributed 9 Talers to the project in total, you and every other player of your group receives $0.4 \times 9 = 3.6$ Talers from the project.

For every Taler you keep for yourself, you earn an income of 1 Taler. If you contribute one Taler of your endowment to the project, the sum of all contributions increases by 1 Taler and your income increases by $0.4 \times 1 = 0.4$ Taler. However, the income of every other member of your group also increases by 0.4 Taler. That is, the total group income increases by $0.4 \times 4 = 1.6$ Talers. Consequently, the other players in your group earn from your contributions to the project and vice versa; you also earn from the contributions of the other members of your group. For each Taler that another member of your group contributes to the project, you earn $0.4 \times 1 = 0.4$ Taler.

The second stage

In the second stage, you learn how much the other players of your group contributed to the project. Additionally, by assigning points, you can **reduce** the income of any other player of your group or **leave it equal**. Also the other players can reduce **your** income if they want to.

Player:	Contribution:	Reduction Points:
1	0	
2	10	
3	5	
You	2	
Continue		

To make this decision, you will see on the screen how many Talers every single player contributed to the project. Thus, you are informed about the identification number of the different players for the current round and their contributions.

For **every** player, you have to decide how many reduction points you would like to assign. You have to make an entry

in any case. If you don't want to affect another player's income, please enter 0.

- Every point that you assign to another player reduces this player's income by 1.75 Talers. For example, if you assign 2 points to another player that player's income is reduced by 3.5 Talers.
- You can assign a maximum of 10 reduction points to each player.
- For every point you assign to another player, your own income is reduced by 1 Taler.

The income of the first two stages (the income is calculated for the first time after the first round) is determined as follows:

$$\begin{aligned} \text{Income of one round} &= \text{income at the end of stage 1} \\ &- 1.75 \times \text{number of received reduction points} \\ &- \text{number of assigned reduction points} \end{aligned}$$

Example

Assume that your income after the first stage was 258 Talers. In the second stage, you were assigned two reduction points by other players. Furthermore, you assigned one reduction point to one of the other players of your group. Your income of 258 Talers is reduced by 2×1.75 points assigned from other players. Additionally, your income is reduced by 1 for the reduction point you assigned to another player. Therefore your round income is $258 - 2 \times 1.75 - 1 = 253.5$ Talers.

Comprehension questions

Please reply to all questions. If anything is unclear please contact the experimenters.

- (1) Every player of your group has an endowment of 20 Talers. None of the players (including you) contributes to the project in the first stage.

What is ...

...the increase in your income due to the first stage?

...the increase in the income of all other players due to the first stage? _____

- (2) Every player of your group has an endowment of 20 Talers. In the first stage you contribute 20 Talers to the project. All other players of your group also contribute 20 Talers.

What is...

... the increase in your income due to the first stage?

... the increase in the income of all other players due to the first stage? _____

- (3) Every player of your group has an endowment of 20 Talers. In sum the other 3 members of your group contribute 30 Talers to the project.

- a What is the increase in your income, if you—in addition to the 30 Talers—contribute 0 Taler to the project?

The increase in your income due to the first stage is

- b What is the increase in your income, if you—in addition to the 30 Talers—contribute 15 Talers to the project?

The increase in your income due to the first stage is

- (4) By how many Talers is your current income reduced if you received the following reduction points?

a 1 point? _____

b 4 points? _____

c 5 points? _____

- (5) Assume you would like to reduce another player's income by 4 Talers. How many reduction points do you have to assign?

You have to assign ____ reduction point/s to reduce this player's income by 4 Talers.

B. Sample instructions for equal punishment, rank info

You are about to take part in an economic experiment. In this experiment you can earn a significant amount of money depending on the choices you make. For this reason, it is essential to read the instructions carefully.

The instructions you obtained are for your private use only. Please note that you are not allowed to communicate with other participants during this experiment. Should you have any questions, please ask us for assistance. If you do not comply with the rules, we will have to exclude you from the experiment and from all payments.

During the experiment, we do not speak of euros but of Talers. Your income will be calculated first in Talers. The total number of Talers collected during the experiment will be converted to euros and rounded to one decimal place according to the following rule:

$$50 \text{ Talers} = 1 \text{ Euro.}$$

At the beginning, every participant receives a one-time lump sum payment of **250 Talers**. With these 250 Talers you can make up for possible losses. The total profit made during the experiment will be paid out in cash at the end of the experiment. Today's experiment consists of several parts. This part of the experiment consists of **10 rounds** in total. Participants are randomly assigned to groups of 4. Thus, apart from you, your group has **3 more players**. For these 10 rounds the composition of your group remains unchanged. Therefore you are in a group with the same persons for 10 rounds. Please note that the identification numbers assigned to you and the other players of your group change

randomly. In this way, individual players cannot be identified over the 10 rounds.

Every round consists of **2 stages**. In the first stage you have to decide how many Talers you want to contribute to a project. In the second stage you are informed about the contributions of all other players and by assigning reduction points, you can decide whether and how much the income of the other players is reduced. On the following pages we describe the exact procedure of the experiment.

Information on the exact procedure of the experiment

The first stage

At the beginning of each round every participant receives **20 Talers**, which we refer to as the player's **endowment** in the following. Your task is to make a decision regarding the use of your endowment. You have to decide how many of the 20 Talers you contribute to a group project and how many of them you keep for yourself. The consequences of this decision will be explained below in more detail.

In this round you receive an endowment of **20 Talers**.
Please decide how much of that endowment you want to contribute.
Your contribution to the group project is:

In each round your **endowment is 20 Talers**. You make a decision about your contribution by entering a number between 0 and 20 in the input field on the screen. By deciding on your contribution, you automatically decide how many Talers you keep for yourself, namely **20 minus your contribution**.

After all players of your group have made their decision, you are informed about the sum of all contributions (including yours) which your group made to the project. Additionally, you are informed about your own contribution, the number of Talers you kept for yourself, as well as your total income from the first stage.

Your income from the first stage consists of two parts:

- (1) The Talers, you kept for yourself ("income from Talers kept")
- (2) and the "income from the project". Your income from the project is 0.4 times the sum of all contributions to the project.

Thus your income from the first stage is calculated as follows:

Income at the end of the first stage :

= Income from Talers kept + Income from the project where

Income from the project = $0.4 \times \text{sum of all contributions to the project}$

The income at the end of the first stage is calculated for all players of your group according to the same formula.

If, for example, the sum of all contributions of the group members is 60 Talers, you and the other players from your group receive the income from the project, equal to $0.4 \times 60 = 24$ Talers. If the players contributed 9 Talers to the project in total, you and every other player of your group receives $0.4 \times 9 = 3.6$ Talers from the project.

For every Taler you keep for yourself, you earn an income of 1 Taler. If you contribute one Taler of your endowment to the project, the sum of all contributions increases by 1 Taler and your income increases by $0.4 \times 1 = 0.4$ Taler. However, the income of every other member of your group also increases by 0.4 Taler. That is, the total group income increases by $0.4 \times 4 = 1.6$ Talers. Consequently, the other players in your group earn from your contributions to the project and vice versa; you also earn from the contributions of the other members of your group. For each Taler that another member of your group contributes to the project, you earn $0.4 \times 1 = 0.4$ Taler.

The second stage

In the second stage, you learn how much the other players of your group contributed to the project. Additionally, you are informed about the order of the current total incomes. For example, if you have the second-highest total income at this point, you would be in **position 2**. Furthermore, by assigning reduction points, you can **reduce** the income of any other player of your group or **leave it equal**. Also the other players can reduce **your** income if they want to.

Player:	Contribution:	Position:	Reduction Points:
1	0	1	
2	10	4	
3	5	3	
You	2	2	
Continue			

To make this decision, you will see on the screen, how many Talers every single player contributed to the project and at what position, determined by the current total income, this player is in. Thus, you are informed about the identification number of the different players for the current round and their contributions and positions.

In case one level of income appears several times, the respective players share one position. Because all players have the same total income in the first round, every player is at **position 1**.

Please note: The total income is determined after completion of one round, because the total income consists of incomes and expenses of both stages. Therefore, you do not have a total income in the first round, but only your lump-sum payment of 250 Talers.

For **every** player, you have to decide how many reduction points you would like to assign. You have to make an entry

in any case. If you don't want to affect another player's income, please enter 0.

- Every point that you assign to another player reduces this player's income by 1.75 Talers.¹³ For example, if you assign 2 points to another player, that player's is reduced by 3.5 Talers.
- You can assign a maximum of 10 reduction points to each player.
- For every point you assign to another player, your own income is reduced by 1 Taler.

The income of the first two stages (the income is calculated for the first time after the first round) is determined as follows:

$$\begin{aligned} \text{Income of one round} &= \text{income at the end of stage 1} \\ &- 1.75 \times \text{number of received reduction points} \\ &- \text{number of assigned reduction points} \end{aligned}$$

Example

Assume that your income after the first stage was 258 Talers. In the second stage, you were assigned two reduction points by other players. Furthermore, you assigned one reduction point to one of the other players of your group. Your income of 258 Talers is reduced by 2×1.75 points assigned from other players. Additionally, your income is reduced by 1 for the reduction point you assigned to another player. Therefore your round income is: $258 - 2 \times 1.75 - 1 = 253.5$ Talers.

Comprehension questions

Please reply to all questions. If anything is unclear please contact the experimenters.

- (1) Every player of your group has an endowment of 20 Talers. None of the players (including you) contributes to the project in the first stage.

What is ...

...the increase in your income due to the first stage?

...the increase in the income of all other players due to the first stage? _____

- (2) Every player of your group has an endowment of 20 Talers. In the first stage you contribute 20 Talers to the project. All other players of your group also contribute 20 Talers.

What is...

... the increase in your income due to the first stage?

¹³In Experiment 1, the punishment effectiveness was 1 instead of 1.75. The instructions for the equal stable condition (with rank info) only differed in this respect.

... the increase in the income of all other players due to the first stage? _____

- (3) Every player of your group has an endowment of 20 Talers. In sum the other 3 members of your group contribute 30 Talers to the project.

- a What is the increase in your income, if you—in addition to the 30 Talers—contribute 0 Taler to the project?

The increase in your income due to the first stage is

- b What is the increase in your income, if you—in addition to the 30 Talers—contribute 15 Talers to the project?

The increase in your income due to the first stage is

- (4) By how many Talers is your current income reduced if you received the following reduction points from other players?

- a 1 point? _____
b 4 points? _____
c 5 points? _____

- (5) Assume you would like to reduce another player's income by 4 Talers. How many reduction points do you have to assign?

You have to assign ____ reduction point/s to reduce this player's income by 4 Talers.

C. Sample instructions unequal, stable punishment

You are about to take part in an economic experiment. In this experiment you can earn a significant amount of money depending on the choices you make. For this reason, it is essential to read the instructions carefully.

The instructions you obtained are for your private use only. Please note that you are not allowed to communicate with other participants during this experiment. Should you have any questions, please ask us for assistance. If you do not comply with the rules, we will have to exclude you from the experiment and from all payments.

During the experiment, we do not speak of euros but of Talers. Your income will be calculated first in Talers. The total number of Talers collected during the experiment will be converted to euros and rounded to one decimal place according to the following rule:

$$50 \text{ Talers} = 1 \text{ Euro.}$$

At the beginning, every participant receives a one-time lump sum payment of **250 Talers**. With these 250 Talers you can make up for possible losses. The total profit made during the experiment will be paid out in cash at the end of the experiment. Today's experiment consists of several parts.

This part of the experiment consists of **10 rounds** in total. Participants are randomly assigned to groups of 4. Thus, apart from you, your group has **3 more players**. For these 10 rounds the composition of your group remains unchanged. Therefore you are in a group with the same persons for 10 rounds. One group consists of **1 type A player** and **3 type B players**. Please note that the identification numbers assigned to you and the other players of your group change randomly. In this way, individual players cannot be identified over the 10 rounds.

Every round consists of **2 stages**. In the first stage you have to decide how many Talers you want to contribute to a project. In the second stage you are informed about the contributions of all other players and by assigning reduction points, you can decide whether and how much the income of the other players is reduced. On the following pages we describe the exact procedure of the experiment.

Information on the exact procedure of the experiment

The first stage

At the beginning of each round every participant receives **20 Talers**, which we refer to as the player's **endowment** in the following. Your task is to make a decision regarding the use of your endowment. You have to decide how many of the 20 Talers you contribute to a group project and how many of them you keep for yourself. The consequences of this decision will be explained below in more detail.

In this round you receive an endowment of **20 Talers**.
Please decide how much of that endowment you want to contribute.
Your contribution to the group project is:

Continue

In each round your **endowment** is **20 Talers**. You make a decision about your contribution by entering a number between 0 and 20 in the input field on the screen. By deciding on your contribution you automatically decide how many Talers you keep for yourself, namely **20 minus your contribution**.

After all players of your group have made their decision, you are informed about the sum of all contributions (including yours) which your group made to the project. Additionally, you are informed about your own contribution, the number of Talers you kept for yourself, as well as your total income from the first stage.

Your income from the first stage consists of two parts:

- (1) The Talers, you kept for yourself ("income from Talers kept")
- (2) and the "income from the project". Your income from the project is 0.4 times the sum of all contributions to the project.

Thus your income from the first stage is calculated as follows:

Income at the end of the first stage :

= Income from Talers kept + Income from the project where
Income from the project = $0.4 \times \text{sum of all contributions to the project}$.

The income at the end of the first stage is calculated for all players of your group according to the same formula.

If, for example, the sum of all contributions of the group members is 60 Talers, you and the other players from your group receive the income from the project, equal to $0.4 \times 60 = 24$ Talers. If the players contributed 9 Talers to the project in total, you and every other player of your group receives $0.4 \times 9 = 3.6$ Talers from the project.

For every Taler you keep for yourself, you earn an income from 1 Taler. If you contribute one Taler of your endowment to the project, the sum of all contributions increases by 1 Taler and your income increases by $0.4 \times 1 = 0.4$ Taler. However, the income of every other member of your group also increases by 0.4 Taler. That is, the total group income increases by $0.4 \times 4 = 1.6$ Talers. Consequently, the other players in your group earn from your contributions to the project and vice versa; you also earn from the contributions of the other members of your group. For each Taler that another member of your group contributes to the project, you earn $0.4 \times 1 = 0.4$ Taler.

The second stage

In the second stage, you learn how much the other players of your group contributed to the project. Additionally, you are informed about the order of the current total incomes. For example, if you have the second-highest total income at this point, you would be in **position 2**. Furthermore, by assigning reduction points, you can **reduce** the income of any other player of your group or **leave it equal**. Also the other players can reduce **your** income if they want to.

In this round you are player type A!

Player:	Contribution:	Position:	Reduction Points:
1	0	1	
2	10	4	
3	5	3	
You	2	2	

Continue

To make this decision, you will see on the screen how many Talers every single player contributed to the project and at what position, determined by the current total income, this player is in. Thus, you are informed about the identification number of the different players for the current round and

their contributions and positions. Furthermore, you learn which player type you are.

In case one level of income appears several times, the respective players share one position. Because all players have the same total income in the first round, every player is at **position 1**.

Please note: The total income is determined after completion of one round, because the total income consists of incomes and expenses of both stages. Therefore, you do not have a total income in the first round, but only your lump-sum payment of 250 Talers.

For **every** player, you have to decide how many reduction points you would like to assign. You have to make an entry in any case. If you don't want to affect another player's income, please enter 0.

The opportunity to assign reduction points depends on whether you are a **type A player** or a **type B player**.

- Every point that **type A player** assigns to another player reduces this player's income by 4 Talers. For example, if a type A player assigns 2 points to you, your income is reduced by 8 Talers.
- Every point that one of the three **type B players** assigns to another player reduces this player's income by 1 Taler. For example, if a type B player assigns 2 reduction points to you, your income is reduced by 2 Talers.
- In case you are a type A player, you can assign a maximum of 5 reduction points to each player, in case you are a type B player, you can assign a maximum of 10 reduction points to each player.
- For every point you assign to another player, your own income is reduced by 1 Taler whether you are type A player or type B player.

Prior to the first round you are randomly assigned a player role. You keep this player role for all 10 rounds.

The income of the first two stages (the income is calculated for the first time after the first round) is determined as follows, depending on the player type:

Type A player

Income of one round = income at the end of stage 1

- number of received reduction points
- number of assigned reduction points

Type B player

Income of one round = income at the end of stage 1

- $4 \times (\text{number of reduction points received from type A player})$
- number of reduction points received from type B players
- number of assigned reduction points

Example

Assume that you are a type B player and your income after the first stage was 258 Talers. In the second stage you were assigned one reduction point by type A player and one reduction point by type B players. Furthermore, you assigned one point to one of the other players of your group. Your income of 258 Talers is reduced by 4×1 point assigned from type A player and 1 point from type B player. Additionally, your income is reduced by 1 for the reduction point you assigned to

another player. Therefore your round income is:
 $258 - 4 - 1 - 1 = 252$ Talers.

Comprehension questions

Please reply to all questions. If anything is unclear please contact the experimenters.

- (1) Every player of your group has an endowment of 20 Talers. None of the players (including you) contributes to the project in the first stage.

What is ...

...the increase in your income due to the first stage?

...the increase in the income of all other players due to the first stage? _____

- (2) Every player of your group has an endowment of 20 Talers. In the first stage you contribute 20 Talers to the project. All other players of your group also contribute 20 Talers.

What is...

... the increase in your income due to the first stage?

... the increase in the income of all other players due to the first stage? _____

- (3) Every player of your group has an endowment of 20 Talers. In sum the other 3 members of your group contribute 30 Talers to the project.

- a What is the increase in your income, if you—in addition to the 30 Talers—contribute 0 Taler to the project?

The increase in your income due to the first stage is

- b What is the increase in your income, if you—in addition to the 30 Talers—contribute 15 Talers to the project?

The increase in your income due to the first stage is

- (4) By how many Talers is your current income reduced if you received the following reduction points from type A player?

a 1 point? _____

b 4 points? _____

c 5 points? _____

- (5) By how many Talers is your current income reduced if you received the following reduction points from type B player?

a 1 point? _____

- b 4 points? _____
 c 10 points? _____

(6) What is the probability that you are type A player in the second round, if ...

- a ...all players have 240 Talers after the first round?
 b You are type A player in the second round...
 c o with a probability of 100% o with a probability of 25%
 d o with a probability of 50% o with a probability of 0%
 e ...you and player 1 both have 272 Talers and players 3 and 4 both have 271 Talers?

You are type A player in the second round...

o with a probability of 100% o with a probability of 25%

o with a probability of 50% o with a probability of 0%

- f ...you have 273.8 Talers, players 2 and 3 both have 270.8 Talers and player 4 has 268.8 Talers?

You are type A player in the second round...

o with a probability of 100% o with a probability of 25%

o with a probability of 50% o with a probability of 0%

- (7) Assume that you are type A player and you would like to reduce another player's income by 4 Talers. How many reduction points do you have to assign?

You have to assign ____ reduction point/s to reduce this player's income by 4 Talers.

D. Sample instructions unequal, stable punishment (social)

You are about to take part in an economic experiment. In this experiment you can earn a significant amount of money depending on the choices you make. For this reason, it is essential to read the instructions carefully.

The instructions you obtained are for your private use only. Please note that you are not allowed to communicate with other participants during this experiment. Should you have any questions, please ask us for assistance. If you do not comply with the rules, we will have to exclude you from the experiment and from all payments.

During the experiment, we do not speak of euros but of Talers. Your income will be calculated first in Talers. The total number of Talers collected during the experiment will be converted to euros and rounded to one decimal place according to the following rule:

$$50 \text{ Talers} = 1 \text{ Euro.}$$

At the beginning, every participant receives a one-time lump sum payment of **250 Talers**. With these 250 Talers you can make up for possible losses. The total profit made during the experiment will be paid out in cash at the end of

the experiment. Today's experiment consists of several parts. This part of the experiment consists of **10 rounds** in total. Participants are randomly assigned to groups of 4. Thus, apart from you, your group has **3 more players**. For these 10 rounds the composition of your group remains unchanged. Therefore you are in a group with the same persons for 10 rounds. One group consists of **1 type A players** and **3 type B players**. Please note that the identification numbers assigned to you and the other players of your group change randomly. In this way, individual players cannot be identified over the 10 rounds.

Every round consists of **2 stages**. In the first stage you have to decide how many Talers you want to contribute to a project. In the second stage you are informed about the contributions of all other players and by assigning reduction points you can decide whether and how much the income of the other players is reduced. On the following pages we describe the exact procedure of the experiment.

Information on the exact procedure of the experiment

The first stage

At the beginning of each round every participant receives **20 Talers**, which we refer to as the player's **endowment** in the following. Your task is to make a decision regarding the use of your endowment. You have to decide how many of the 20 Talers you contribute to a group project and how many of them you keep for yourself. The consequences of this decision will be explained below in more detail.

In this round you receive an endowment of **20 Talers**.
 Please decide how much of that endowment you want to contribute.
 Your contribution to the group project is:

In each round your **endowment** is **20 Talers**. You make a decision about your contribution by entering a number between 0 and 20 in the input field on the screen. By deciding on your contribution, you automatically decide how many Talers you keep for yourself, namely **20 minus your contribution**.

After all players of your group have made their decision, you are informed about the sum of all contributions (including yours) which your group made to the project. Additionally, you are informed about your own contribution, the number of Talers you kept for yourself, as well as your total income from the first stage.

Your income from the first stage consists of two parts:

- (1) The Talers, you kept for yourself ("income from Talers kept")
- (2) and the "income from the project". Your income from the project is 0.4 times the sum of all contributions to the project.

Thus your income from the first stage is calculated as follows:

Income at the end of the first stage :

= Income from Talers kept + Income from the project where

Income from the project = $0.4 \times \text{sum of all contributions to the project}$.

The income at the end of the first stage is calculated for all players of your group according to the same formula.

If, for example, the sum of all contributions of the group members is 60 Talers, you and the other players from your group receive the income from the project, equal to $0.4 \times 60 = 24$ Talers. If the players contributed 9 Talers to the project in total, you and every other player of your group receives $0.4 \times 9 = 3.6$ Talers from the project.

For every Taler you keep for yourself, you earn an income of 1 Taler. If you contribute one Taler of your endowment to the project, the sum of all contributions increases by 1 Taler and your income increases by $0.4 \times 1 = 0.4$ Taler. However, the income of every other member of your group also increases by 0.4 Taler. That is, the total group income increases by $0.4 \times 4 = 1.6$ Talers. Consequently, the other players in your group earn from your contributions to the project and vice versa; you also earn from the contributions of the other members of your group. For each Taler that another member of your group contributes to the project, you earn $0.4 \times 1 = 0.4$ Taler.

The second stage

In the second stage, you learn how much the other players of your group contributed to the project. Additionally, you are informed about the order of positions which are determined by the current total contributions (to the project) minus received reduction points. For example, if you have the second-highest sum of total contribution minus reduction points at this point, you would be in **position 2**. Furthermore, by assigning reduction points, you can **reduce** the income of any other player of your group or **leave it equal**. Also the other players can reduce **your** income if they want to.

In this round you are player type A!

Player:	Contribution:	Position:	Reduction Points:
1	0	1	
2	10	4	
3	5	3	
You	2	2	

To make this decision, you will see on the screen how many Talers every single player contributed to the project and at what position this player is. Thus, you are informed

about the identification number of the different players for the current round and their contributions and positions. Furthermore, you learn which player type you are.

In case one level of total contributions minus received punishments appears several times, the respective players share one position. Because this amount is calculated for the first time after the first round, every player is at **position 1** in the first round.

For **every** player you have to decide how many reduction points you would like to assign. You have to make an entry in any case. If you don't want to affect another player's income, please enter 0.

The opportunity to assign reduction points depends on whether you are a **type A player** or a **type B player**.

- Every point that **type A player** assigns to another player reduces this player's income by 4 Talers. For example, if type A player assigns 2 points to you, your income is reduced by 8 Talers.
- Every point that one of the three **type B players** assigns to another player, reduces this player's income by 1 Taler. For example, if type B player assigns 2 points to you, your income is reduced by 2 Talers.
- In case you are type A player you can assign a maximum of 5 reduction points to each player, in case you are type B player, you can assign a maximum of 10 reduction points to each player.
- For every point you assign to another player, your income is reduced by 1 Taler whether you are type A player or type B player.

Prior to the first round you are randomly assigned a player role. You keep this role for all 10 rounds.

The income of the first two stages (the income is calculated for the first time after the first round) is determined as follows, depending on the player type:

Type A player

Income of one round = income at the end of stage 1

- number of received reduction points
- number of assigned reduction points

Type B player

Income of one round = income at the end of stage 1

- $4 \times (\text{number of reduction points received from type A player})$
- number of reduction points received from type B players
- number of assigned reduction points

Example:

Assume that you are type B player and your income after the first stage was 258 Talers. In the second stage, you were assigned one reduction point by type A player and one reduction point by one of the type B players. Furthermore, you assigned one point to one of the other players of your group. Your income of 258 Talers is reduced by 4×1 point assigned from type A player and 1 point from type B player. Additionally, your income is reduced by 1 for the reduction point you assigned to another player. Therefore your round income is: $258 - 4 - 1 - 1 = 252$ Talers.

Comprehension questions

Please reply to all questions. If anything is unclear please contact the experimenters.

- (1) Every player of your group has an endowment of 20 Talers. None of the players (including you) contributes to the project in the first stage.

What is ...

...the increase in your income due to the first stage?

...the increase in the income of all other players due to the first stage? _____

- (2) Every player of your group has an endowment of 20 Talers. In the first stage you contribute 20 Talers to the project. All other players of your group also contribute 20 Talers.

What is...

... the increase in your income due to the first stage?

... the increase in the income of all other players due to the first stage? _____

- (3) Every player of your group has an endowment of 20 Talers. In sum the other 3 members of your group contribute 30 Talers to the project.

- a What is the increase in your income, if you—in addition to the 30 Talers—contribute 0 Taler to the project?

The increase in your income due to the first stage is

- b What is the increase in your income, if you—in addition to the 30 Talers—contribute 15 Talers to the project?

The increase in your income due to the first stage is

- (4) By how many Talers is your current income reduced if you received the following reduction points from type A player?

- a 1 point? _____
b 4 points? _____
c 5 points? _____

- (5) By how many Talers is your current income reduced if you received the following reduction points from type B player?

- a 1 point? _____
b 4 points? _____
c 10 points? _____

- (6) Assume you are type A player and you would like to reduce another player's income by 4 Talers. How many reduction points do you have to assign?

You have to assign ____ reduction point/s to reduce this player's income by 4 Talers.

E. Sample instructions unequal, unstable punishment

You are about to take part in an economic experiment. In this experiment you can earn a significant amount of money depending on the choices you make. For this reason, it is essential to read the instructions carefully.

The instructions you obtained are for your private use only. Please note that you are not allowed to communicate with other participants during this experiment. Should you have any questions, please ask us for assistance. If you do not comply with the rules, we will have to exclude you from the experiment and from all payments.

During the experiment, we do not speak of euros but of Talers. Your income will be calculated first in Talers. The total number of Talers collected during the experiment will be converted to euros and rounded to one decimal place according to the following rule:

$$50 \text{ Talers} = 1 \text{ Euro.}$$

At the beginning, every participant receives a one-time lump sum payment of **250 Talers**. With these 250 Talers you can make up for possible losses. The total profit made during the experiment will be paid out in cash at the end of the experiment. Today's experiment consists of several parts. This part of the experiment consists of **10 rounds** in total. Participants are randomly assigned to groups of 4. Thus, apart from you, your group has **3 more players**. For these 10 rounds the composition of your group remains unchanged. Therefore you are in a group with the same persons for 10 rounds. One group consists of **1 type A** player and **3 type B** players. Please note that the identification numbers assigned to you and the other players of your group change randomly. In this way, individual players cannot be identified over the 10 rounds.

Every round consists of **2 stages**. In the first stage you have to decide how many Talers you want to contribute to a project. In the second stage you are informed about the contributions of all other players and by assigning reduction points, you can decide whether and how much the income of the other players is reduced. On the following pages we describe the exact procedure of the experiment.

*Information on the exact procedure of the experiment**The first stage*

At the beginning of each round every participant receives **20 Talers**, which we refer to as the player's **endowment** in the following. Your task is to make a decision regarding the

use of your endowment. You have to decide how many of the 20 Talers you contribute to a group project and how many of them you keep for yourself. The consequences of this decision will be explained below in more detail.

In this round you receive an endowment of 20 Talers.
Please decide how much of that endowment you want to contribute.
Your contribution to the group project is:

In each round your **endowment** is 20 Talers. You make a decision about your contribution by entering a number between 0 and 20 in the input field on the screen. By deciding on your contribution, you automatically decide how many Talers you keep for yourself, namely **20 minus your contribution**.

After all players of your group have made their decision, you are informed about the sum of all contributions (including yours) which your group made to the project. Additionally, you are informed about your own contribution, the number of Talers you kept for yourself, as well as your total income from the first stage.

Your income from the first stage consists of two parts:

- (1) The Talers, you kept for yourself (“income from Talers kept”)
- (2) and the “income from the project”. Your income from the project is 0.4 times the sum of all contributions to the project.

Thus your income from the first stage is calculated as follows:

Income at the end of the first stage :

= Income from Talers kept + Income from the project
where

Income from the project = $0.4 \times \text{sum of all contributions to the project}$.

The income at the end of the first stage is calculated for all players of your group according to the same formula. If, for example, the sum of all contributions of the group members is 60 Talers, you and the other players from your group receive the income from the project, equal to $0.4 \times 60 = 24$ Talers. If the players contributed 9 Talers to the project in total, you and every other player of your group receives $0.4 \times 9 = 3.6$ Talers from the project.

For every Taler you keep for yourself, you earn an income of 1 Taler. If you contribute one Taler of your endowment to the project, the sum of all contributions increases by 1 Taler and your income increases by $0.4 \times 1 = 0.4$ Taler. However, the income of every other member of your group also increases by 0.4 Taler. That is, the total group income increases by $0.4 \times 4 = 1.6$ Talers. Consequently, the other players in your group earn from your contributions to the project and vice versa; you also earn from the contributions of the other members of your group. For each Taler that another member of your group contributes to the project, you earn $0.4 \times 1 = 0.4$ Taler.

The second stage

In the second stage, you learn how much the other players of your group contributed to the project. Additionally, you are informed about the order of the current total incomes. For example, if you have the second-highest total income at this point, you would be in **position 2**. Furthermore, by assigning reduction points, you can **reduce** the income of any other player of your group or **leave it equal**. Also the other players can reduce **your** income if they want to.

In this round you are player type A!

Player:	Contribution:	Position:	Reduction Points:
1	0	1	
2	10	4	
3	5	3	
You	2	2	

To make this decision, you will see on the screen how many Talers every single player contributed to the project and at what position, determined by the current total income, this player is in. Thus, you are informed about the identification number of the different players for the current round and their contributions and positions. Furthermore, you learn which player type you are.

In case one level of income appears several times, the respective players share one position. Because all players have the same total income in the first round, every player is at **position 1**.

Please note: The total income is determined after completion of one round, because the total income consists of incomes and expenses of both stages. Therefore, you do not have a total income in the first round, but only your lump-sum payment of 250 Talers.

For **every** player, you have to decide how many reduction points you would like to assign. You have to make an entry in any case. If you don't want to affect another player's income, please enter 0.

The opportunity to assign reduction points depends on whether you are a **type A player** or a **type B player**.

- Every point, that **type A player** assigns to another player reduces this player's income by 4 Talers. For example, if type A player assigns 2 points to you, your income is reduced by 8 Talers.
- Every point, that one of the three **type B players** assigns to another player, reduces this player's income by 1 Taler. For example, if type B player assigns 2 points to you, your income is reduced by 2 Talers.
- In case you are type A player, you can assign a maximum of 5 reduction points to each player, in case you are type B player, you can assign a maximum of 10 reduction points to each player.

- For every point you assign to another player, your own income is reduced by 1 Taler whether you are type A player or type B player.

Prior to the first round you are randomly assigned a player role. From the second round onwards, always the player with the highest total income is **type A player**. That is, the person who earned the most so far is type A player and can reduce another player's income by 4 by assigning one reduction point.

Your player type can thus change from round to round, depending on the decisions you and the other players of your group make during the experiment. In case one level of total income exists multiple times (more than one player has the highest income), chance determines who of these players receives player role A.

The income of the first two stages (the income is calculated for the first time after the first round) is determined as follows, depending on the player type:

Type A player

- Income of one round = income at the end of stage 1
- number of received reduction points
 - number of assigned reduction points

Type B player

- Income of one round = income at the end of stage 1
- 4*(number of reduction points received from type A player)
 - number of reduction points received from type B players
 - number of assigned reduction points

Example

Assume that you are type B player and your income after the first stage was 258 Talers. In the second stage, you were assigned one reduction point by type A player and one reduction point by one of the type B players. Furthermore, you assigned one point to one of the other players of your group. Your income of 258 Talers is reduced by 4×1 point assigned from type A player and 1 point from type B player. Additionally, your income is reduced by 1 for the reduction point you assigned to another player. Therefore your round income is: $258 - 4 - 1 - 1 = 252$ Talers.

Comprehension questions

Please reply to all questions. If anything is unclear please contact the experimenters.

- (1) Every player of your group has an endowment of 20 Talers. None of the players (including you) contributes to the project in the first stage.

What is ...

...the increase in your income due to the first stage?

...the increase in the income of all other players due to the first stage? _____

- (2) Every player of your group has an endowment of 20 Talers. In the first stage you contribute 20 Talers to the project. All other players of your group also contribute 20 Talers.

What is...

... the increase in your income due to the first stage?

... the increase in the income of all other players due to the first stage? _____

- (3) Every player of your group has an endowment of 20 Talers. In sum the other 3 members of your group contribute 30 Talers to the project.

- a What is the increase in your income, if you—in addition to the 30 Talers—contribute 0 Taler to the project?

The increase in your income due to the first stage is _____

- b What is the increase in your income, if you—in addition to the 30 Talers—contribute 15 Talers to the project?

The increase in your income due to the first stage is _____

- (4) By how many Talers is your current income reduced if you received the following reduction points from type A player?

- a 1 point? _____
- b 4 points? _____
- c 5 points? _____

- (5) By how many Talers is your current income reduced if you received the following reduction points from type B player?

- a 1 point? _____
- b 4 points? _____
- c 10 points? _____

- (6) What is the probability that you are type A player in the second round, if ...

- a ...all players have 240 Talers after the first round?

You are type A player in the second round...

- o with a probability of 100% o with a probability of 25%
- o with a probability of 50% o with a probability of 0%

- b ...you and player 1 both have 272 Talers and players 3 and 4 both have 271 Talers?

You are type A player in the second round...

- o with a probability of 100% o with a probability of 25%

- o with a probability of 50% o with a probability of 0%
- c ...you have 273.8 Talers, players 2 and 3 both have 270.8 Talers and player 4 has 268.8 Talers?

You are type A player in the second round...

- o with a probability of 100% o with a probability of 25%
- o with a probability of 50% o with a probability of 0%
- (7) Assume that you are type A player and you would like to reduce another player's income by 4 Talers. How many reduction points do you have to assign?

You have to assign ____ reduction point/s to reduce this player's income by 4 Talers.

F. Sample instructions unequal, unstable punishment (social)

You are about to take part in an economic experiment. In this experiment you can earn a significant amount of money depending on the choices you make. For this reason, it is essential to read the instructions carefully.

The instructions you obtained are for your private use only. Please note that you are not allowed to communicate with other participants during this experiment. Should you have any questions, please ask us for assistance. If you do not comply with the rules, we will have to exclude you from the experiment and from all payments.

During the experiment, we do not speak of euros but of Talers. Your income will be calculated first in Talers. The total number of Talers collected during the experiment will be converted to euros and rounded to one decimal place according to the following rule:

$$50 \text{ Talers} = 1 \text{ Euro}$$

At the beginning, every participant receives a one-time lump sum payment of **250 Talers**. With these 250 Talers you can make up for possible losses. The total profit made during the experiment will be paid out in cash at the end of the experiment. Today's experiment consists of several parts. This part of the experiment consists of **10 rounds** in total. Participants are randomly assigned to groups of 4. Thus, apart from you, your group has **3 more players**. For these 10 rounds the composition of your group remains unchanged. Therefore you are in a group with the same persons for 10 rounds. One group consists of **1 type A player** and **3 B players**. Please note that the identification numbers assigned to you and the other players of your group change randomly. In this way, individual players cannot be identified over the 10 rounds.

Every round consists of **2 stages**. In the first stage you have to decide how many Talers you want to contribute to a project. In the second stage you are informed about the contributions of all other players and by assigning reduction

points, you can decide whether and how much the income of the other players is reduced. On the following pages we describe the exact procedure of the experiment.

Information on the exact procedure of the experiment

The first stage. At the beginning of each round every participant receives **20 Talers**, which we refer to as the player's **endowment** in the following. Your task is to make a decision regarding the use of your endowment. You have to decide how many of the 20 Talers you contribute to a group project and how many of them you keep for yourself. The consequences of this decision will be explained below in more detail.

In this round you receive an endowment of **20 Talers**. Please decide how much of that endowment you want to contribute.

Your contribution to the group project is:

In each round your **endowment** is **20 Talers**. You make a decision about your contribution by entering a number between 0 and 20 in the input field on the screen. By deciding on your contribution, you automatically decide how many Talers you keep for yourself, namely **20 minus your contribution**.

After all players of your group have made their decision, you are informed about the sum of all contributions (including yours) which your group made to the project. Additionally, you are informed about your own contribution, the number of Talers you kept for yourself, as well as your total income from the first stage.

Your income from the first stage consists of two parts:

- (1) The Talers, you kept for yourself ("income from Talers kept")
- (2) and the "income from the project". Your income from the project is 0.4 times the sum of all contributions to the project.

Thus your income from the first stage is calculated as follows:

Income at the end of the first stage :

= Income from Talers kept + Income from the project where

Income from the project = $0.4 \times \text{sum of all contributions to the project}$

The income at the end of the first stage is calculated for all players of your group according to the same formula.

If, for example, the sum of all contributions of the group members is 60 Talers, you and the other players from your group receive the income from the project, equal to $0.4 \times 60 = 24$ Talers. If the players contributed 9 Talers to the project in total, you and every other player of your group receives $0.4 \times 9 = 3.6$ Talers from the project.

For every Taler you keep for yourself, you earn an income of 1 Taler. If you contribute one Taler of your endowment to the project, the sum of all contributions increases by 1 Taler and your income increases by $0.4 \times 1 = 0.4$ Taler. However, the income of every other member of your group also increases by 0.4 Taler. That is, the total group income increases by $0.4 \times 4 = 1.6$ Talers. Consequently, the other players in your group earn from your contributions to the project and vice versa; you also earn from the contributions of the other members of your group. For each Taler that another member of your group contributes to the project you earn $0.4 \times 1 = 0.4$ Taler.

The second stage. In the second stage, you learn how much the other players of your group contributed to the project. Additionally, you are informed about the order of positions which are determined by the current total contributions (to the project) minus received reduction points. For example, if you have the second-highest sum of total contribution minus reduction points at this point, you would be in **position 2**. Furthermore, by assigning reduction points, you can **reduce** the income of any other player of your group or **leave it equal**. Also the other players can reduce **your** income if they want to.

In this round you are player type A!

Player:	Contribution:	Position:	Reduction Points:
1	0	1	
2	10	4	
3	5	3	
You	2	2	

To make this decision, you will see on the screen how many Talers every single player contributed to the project and at what position this player is in. Thus, you are informed about the identification number of the different players for the current round and their contributions and positions. Furthermore, you learn which player type you are.

In case one level of total contributions minus received punishments appears several times, the respective players share one position. Because this amount is calculated for the first time after the first round, every player is at **position 1** in the first round.

For **every** player, you have to decide how many reduction points you would like to assign. You have to make an entry in any case. If you don't want to affect another player's income, please enter 0.

The opportunity to assign reduction points depends on whether you are **type A player** or **type B player**.

- Every point, that **type A player** assigns to another player reduces this player's income by 4 Talers. For example, if type A player assigns 2 points to you, your income is reduced by 8 Talers.

- Every point, that one of the three **type B players** assigns to another player, reduces this player's income by 1 Taler. For example, if type B player assigns 2 points to you, your income is reduced by 2 Talers.
- In case you are type A player, you can assign a maximum of 5 reduction points to each player, in case you are type B player, you can assign a maximum of 10 reduction points to each player.
- For every point you assign to another player, your own income is reduced by 1 Taler whether you are type A player or type B player.

Prior to the first round you are randomly assigned a player role. From the second round onwards, always the player with the highest sum of total contributions minus received reduction points is **type A player**. That is, the person who has the highest sum of total contributions—received reduction points is type A player and can reduce another player's income by 4 by assigning one reduction point. Your player type can thus change from round to round, depending on the decisions you and the other players of your group make during the experiment. In case one amount exists multiple times (more than one player has the highest amount), chance determines who of these player receives player role A.

The income of the first two stages (the income is calculated for the first time after the first round) is determined as follows, depending on the player type: **Type A player**

- Income of one round = income at the end of stage 1
- number of received reduction points
 - number of assigned reduction points

Type B player

- Income of one round = income at the end of stage 1
- $4 \times (\text{number of reduction points received from type A player})$
 - number of reduction points received from players B
 - number of assigned reduction points

Example

Assume that you are type B player and your income after the first stage was 258 Talers. In the second stage, you were assigned one reduction point by type A player and one reduction point by one of the type B players. Furthermore, you assigned one reduction point to one of the other players of your group. Your income of 258 Talers is reduced by 4×1 point assigned from type A player and 1 point from type B player. Additionally, your income is reduced by 1 for the reduction point you assigned to another player. Therefore your round income is: $258 - 4 - 1 - 1 = 252$ Talers.

Comprehension questions.

Please reply to all questions. If anything is unclear please contact the experimenters.

- (1) Every player of your group has an endowment of 20 Talers. None of the players (including you) contributes to the project in the first stage.

What is ...

...the increase in your income due to the first stage?

...the increase in the income of all other players due to the first stage? _____

- (2) Every player of your group has an endowment of 20 Talers. In the first stage you contribute 20 Talers to the project. All other players of your group also contribute 20 Talers.

What is...

... the increase in your income due to the first stage? _____

... the increase in the income of all other players due to the first stage? _____

- (3) Every player of your group has an endowment of 20 Talers. In sum the other 3 members of your group contribute 30 Talers to the project.

- a What is the increase in your income, if you—in addition to the 30 Talers—contribute 0 Taler to the project?

The increase in your income due to the first stage is _____

- b What is the increase in your income, if you—in addition to the 30 Talers—contribute 15 Talers to the project?

The increase in your income due to the first stage is _____

- (4) By how many Talers is your current income reduced if you received the following reduction points from type A player?

- a 1 point? _____
b 4 points? _____
c 5 points? _____

- (5) By how many Talers is your current income reduced if you received the following reduction points from type B players?

- a 1 point? _____
b 4 points? _____
c 10 points? _____

- (6) What is the probability that you are type A player in the second round, if ...

- a ...all players contributed 15 Talers to the project and received 5 reduction points in the first round?

You are type A player in the second round...

o with a probability of 100% o with a probability of 25%

o with a probability of 50% o with a probability of 0%

- b ...you and player 1 both contributed 15 Talers and players 3 and 4 both contributed 12 Talers and all players received 5 reduction points?

You are type A player in the second round...

o with a probability of 100% o with a probability of 25%

o with a probability of 50% o with a probability of 0%

- c ...all players contributed 15 Talers to the project and you received 5 reduction points, players 2 and 3 received 4 reduction points and player 5 received 3 reduction points?

You are type A player in the second round...

o with a probability of 100% o with a probability of 25%

o with a probability of 50% o with a probability of 0%

- (7) Assume that you are type A player and you would like to reduce another player's income by 4 Talers. How many reduction points do you have to assign?

You have to assign ____ reduction point/s to reduce this player's income by 4 Taler

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